Self Introduction

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Basic Information

Alibaba Group

Sep. 2024 - Present

- Senior Algorithm Engineer in Taobao & Tmall Group
- Responsibilities: Advertisement Reranking System

University of Liverpool

Dec. 2019 - Jul. 2024

- Ph.D. in CS & EE
- Advisors: Prof. Kaizhu Huang, Prof. Xinping Yi

Kaizhu Huang

Professor of Electrical and Computer Engineering at Duke Kunshan University

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Research Projects

Graph Learning Against Homophilous Assumptions

Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review) 2025.*

Jingwei Guo, et.al. ES-GNN: Generalizing Graph Neural Networks Beyond Homophily with Edge Splitting. *TPAMI 2024.*

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW 2023*.

Jingwei Guo, et.al. Learning Disentangled Graph Convolutional Networks Locally and Globally. *TNNLS 2022.*

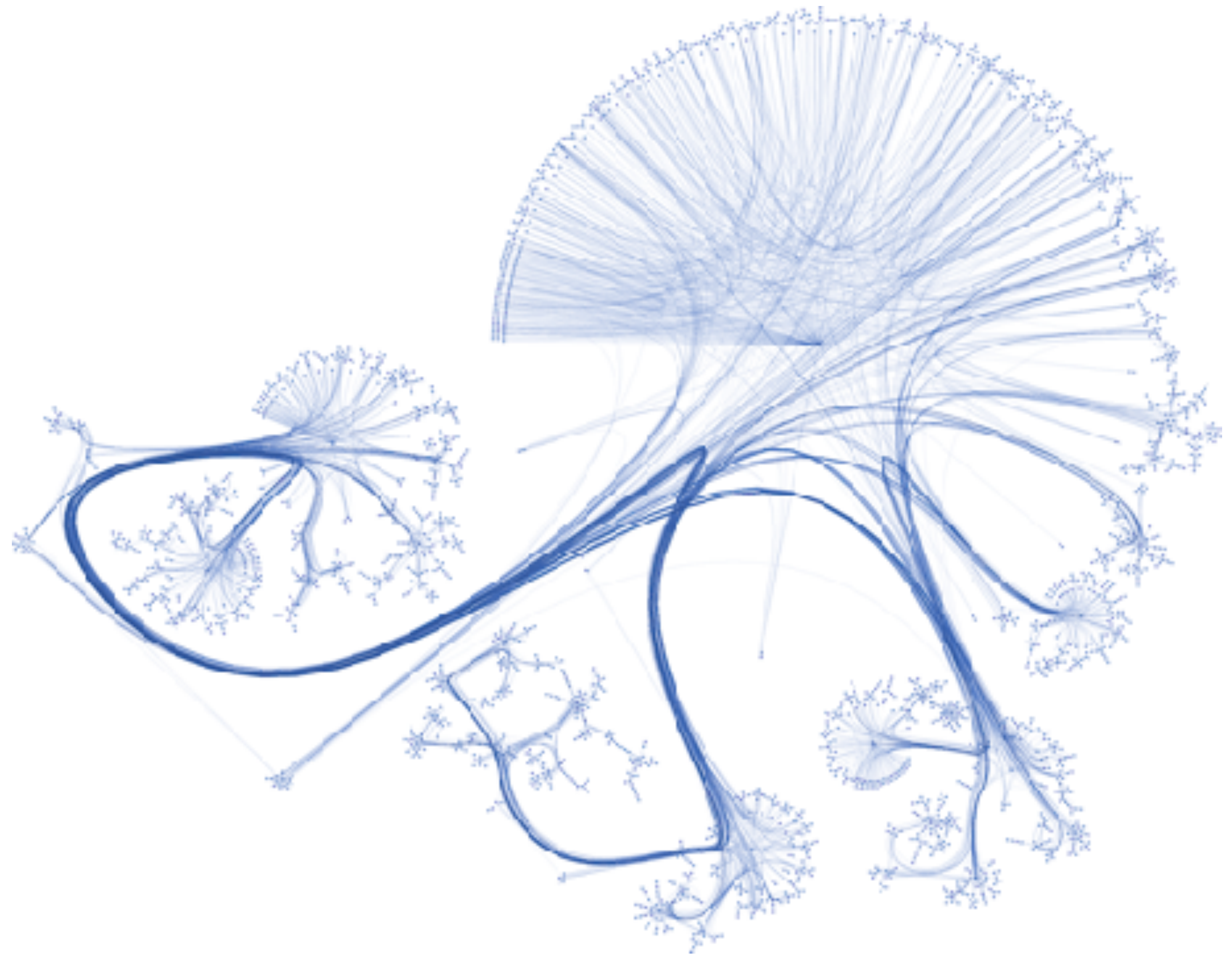
Transfer Learning Under Distribution Shifts

Zixian Su*, **Jingwei Guo***, et.al. Un-mixing Test-time Adaptation under Heterogeneous Data Streams. *TKDE (Under Review) 2025.*

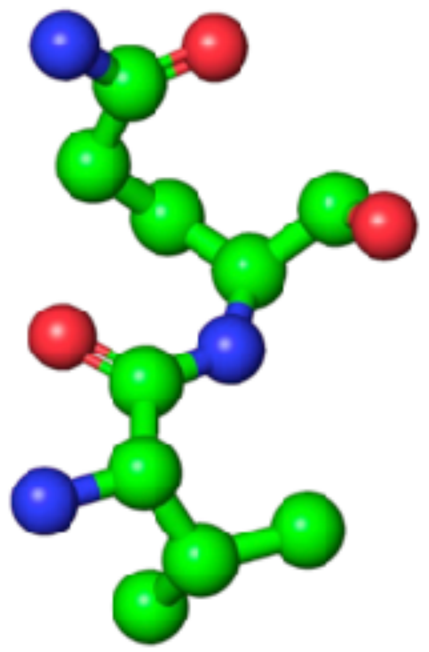
Zixian Su*, **Jingwei Guo***, et.al. Navigating Distribution Shifts in Medical Image Analysis. *ACM Comput. Surv (Under Review) 2025.*

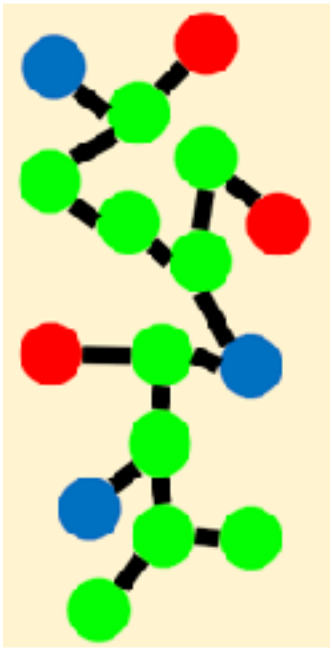
Zixian Su, **Jingwei Guo**, et.al. Unraveling Batch Normalization for Realistic Test-Time Adaptation. *AAAI 2024.*

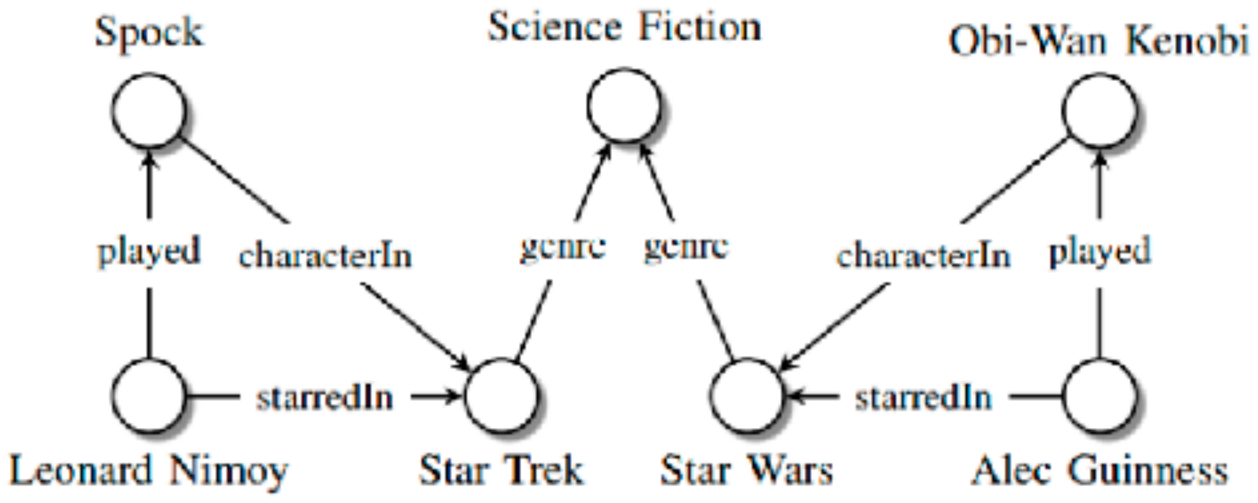
Graph Learning Against Homophilous Assumptions

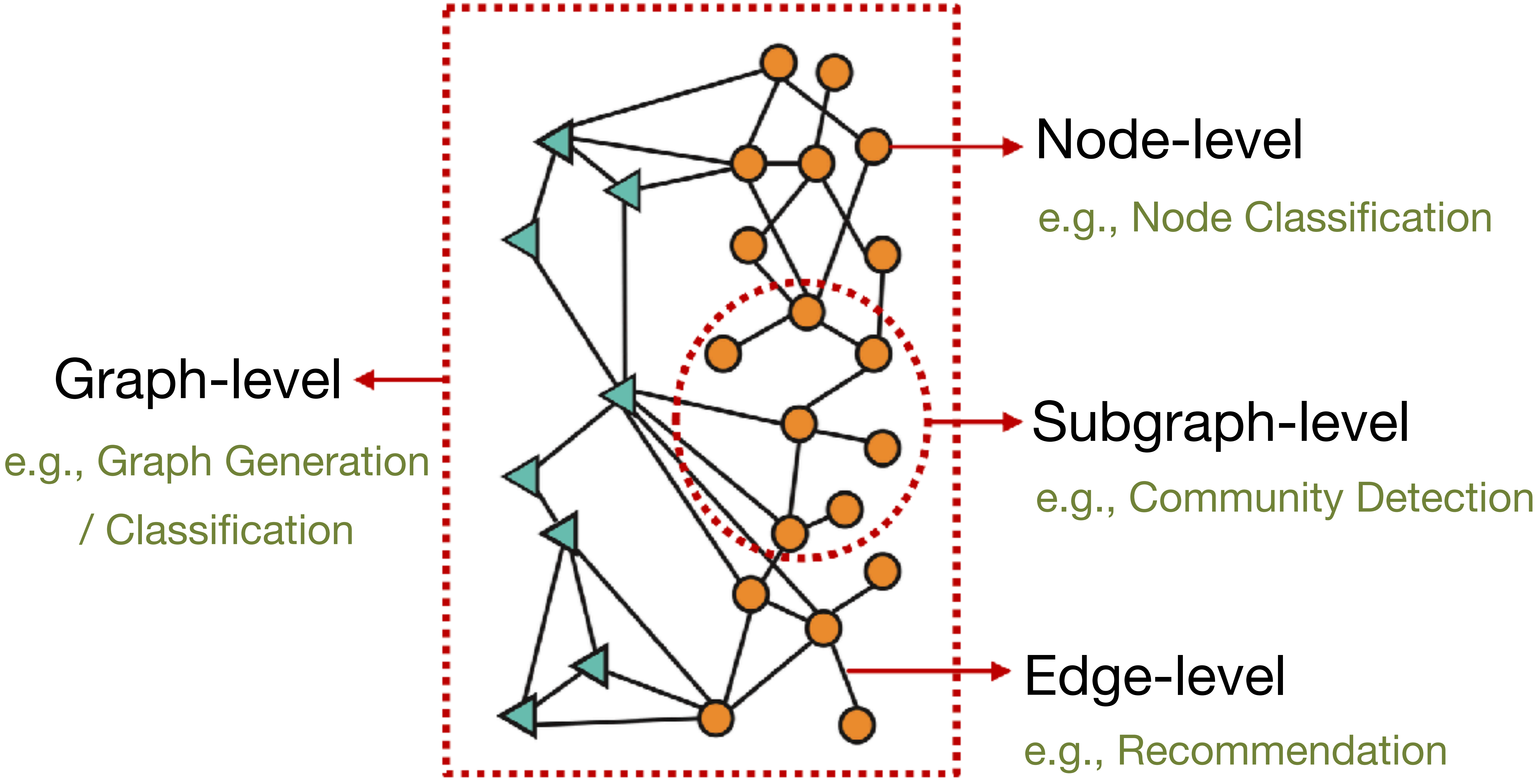


Research Background — Graphs









Graphs are a language describing entities with relations or interactions.

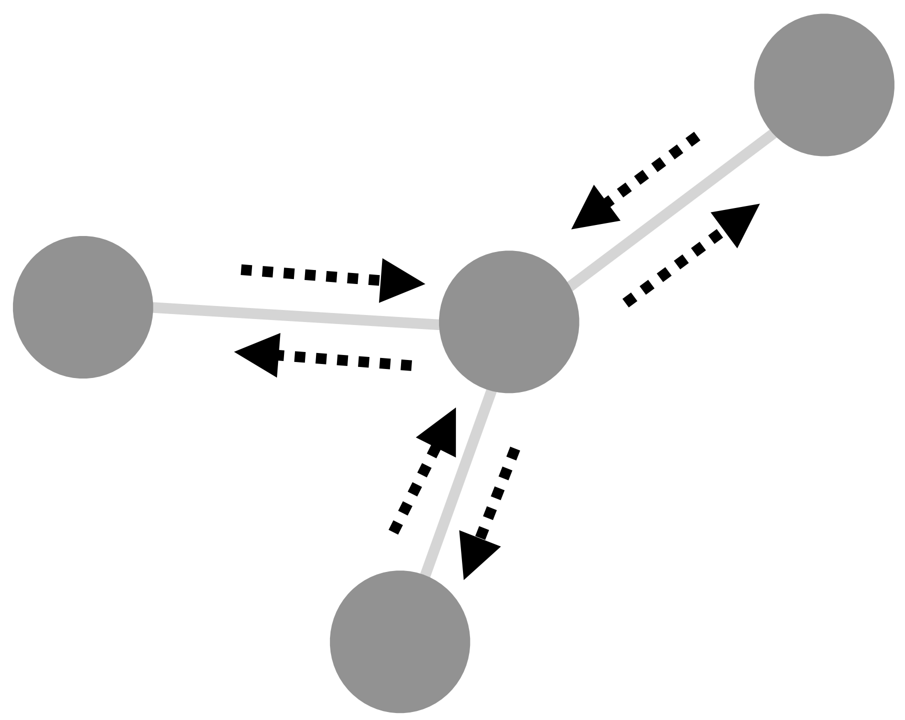
http://cs224w.stanford.edu/

Core Insights

- Integrate both node features and graph topology via either a message passing framework or a spectral filtering operation.

Spatial-based Methods

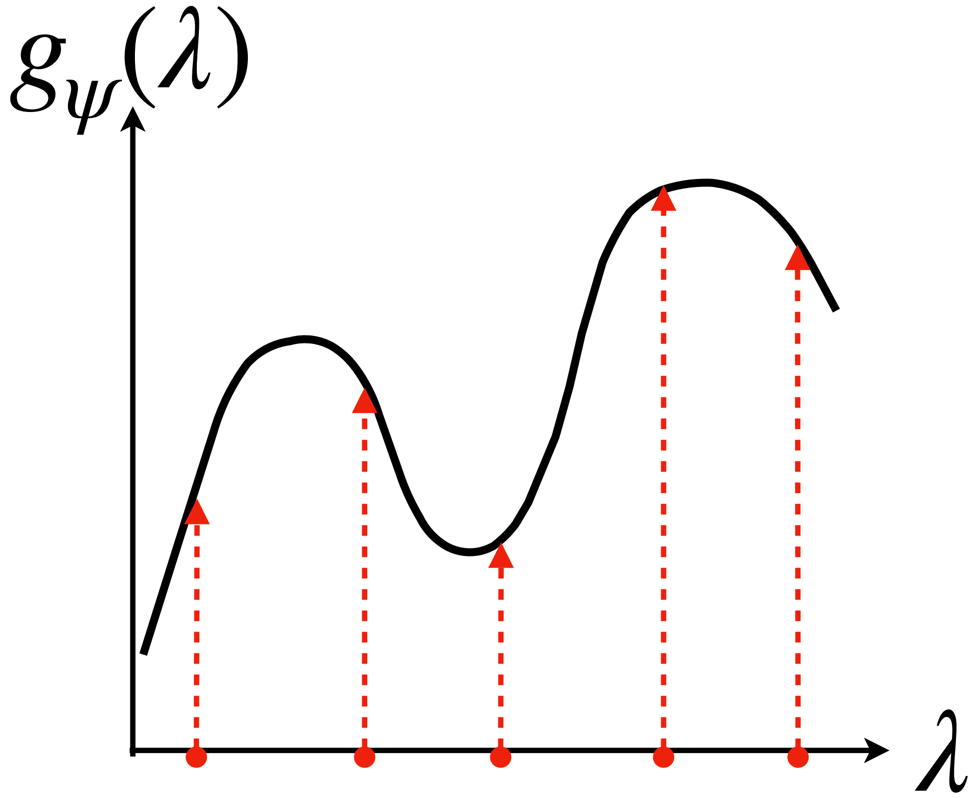
$$\mathbf{z}_i = f_{upd}(\mathbf{x}_i, f_{agg}(\{\mathbf{x}_j \mid \forall v_j \in N_i\}))$$



Message Passing

Spectral-based Methods

$$\mathbf{Z} = \mathbf{U} g_\psi(\mathbf{\Lambda}) \mathbf{U}^T \mathbf{X}$$



Spectral Filtering

Unified Optimization Objective

- Can be interpreted as different solutions to the same graph denoising problem:

$$\arg\min_{\mathbf{Z}} \alpha\|\mathbf{X}-\mathbf{Z}\|_2^2 + (1-\alpha)\mathrm{tr}(\mathbf{Z}^T\hat{\mathbf{L}}\mathbf{Z})$$

$$\sum_{(v_i,v_j)\in E} \|deg_i^{-\frac{1}{2}}\mathbf{z}_i - deg_j^{-\frac{1}{2}}\mathbf{z}_j\|_2^2$$

keep close to the original features

smooth node features across the graph

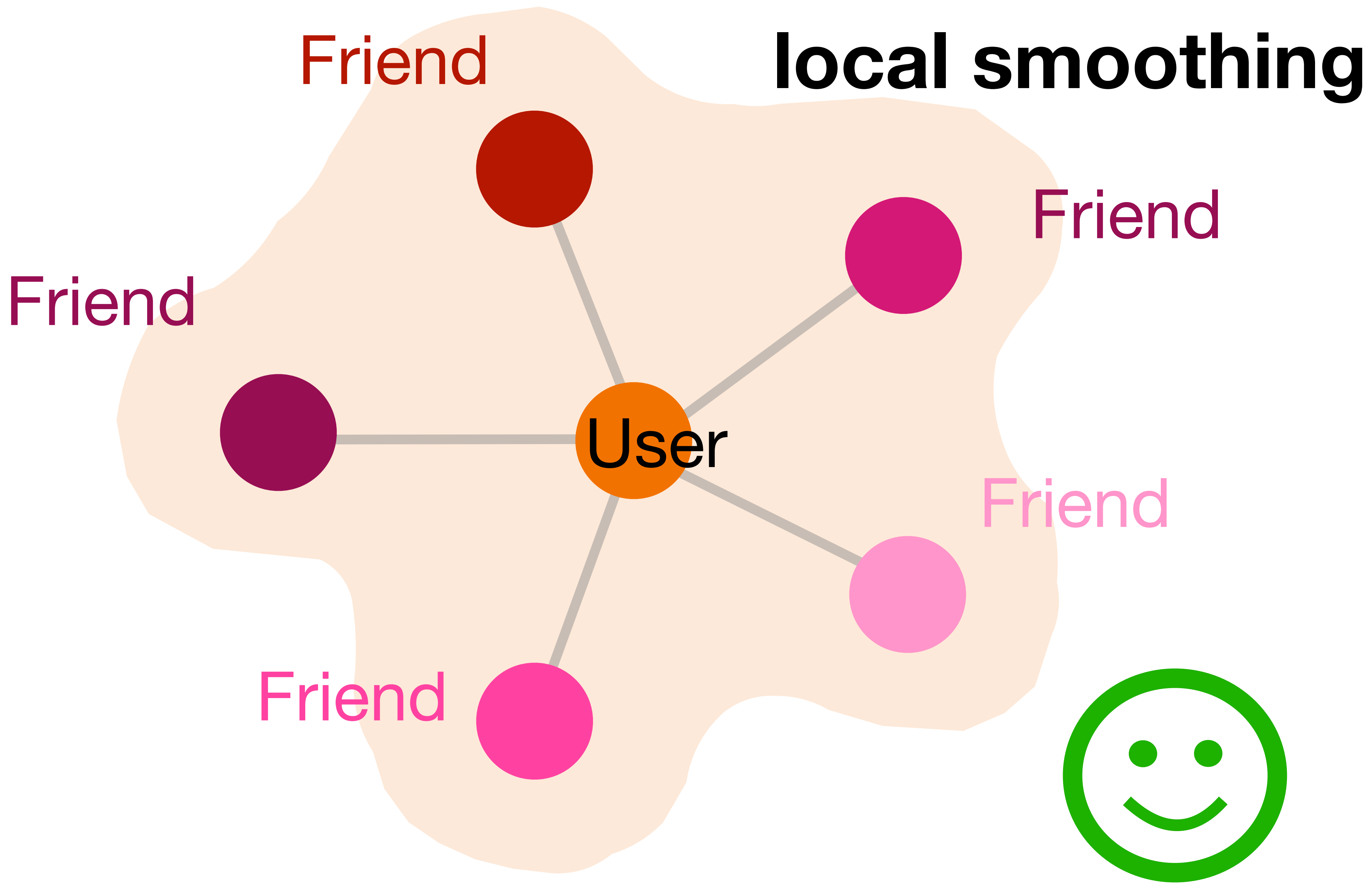
Ma, Y., et.al. A unified view on graph neural networks as graph signal denoising. *CIKM, 2021.* Zhu, M., et.al. Interpreting and unifying graph neural networks with an optimization framework. *WWW, 2021.*

Research Motivation

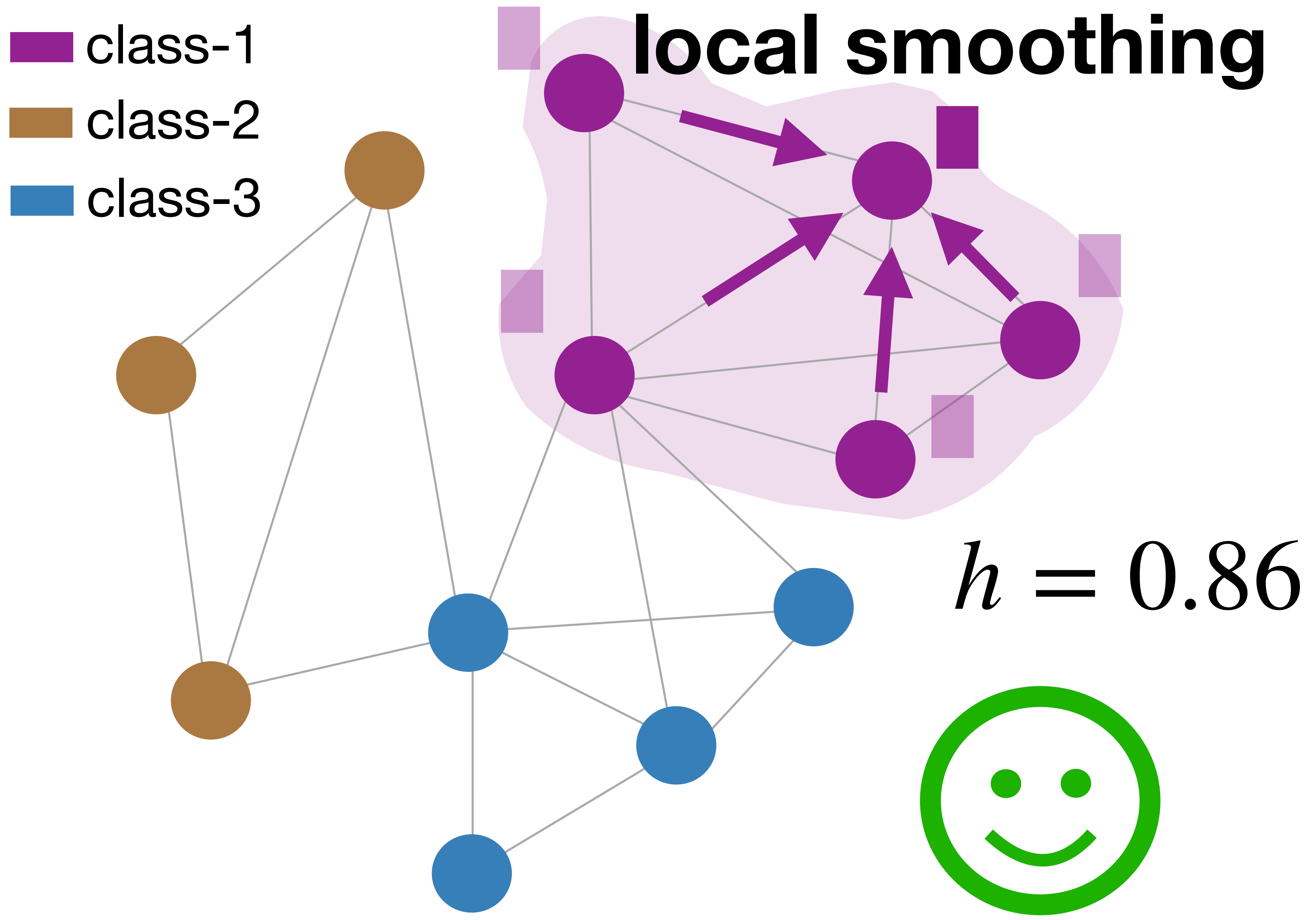
Idealized Scenarios

h = # intra-class edges / # total edges

- Homogenous Node Relationships



- Homophilic Linking Patterns



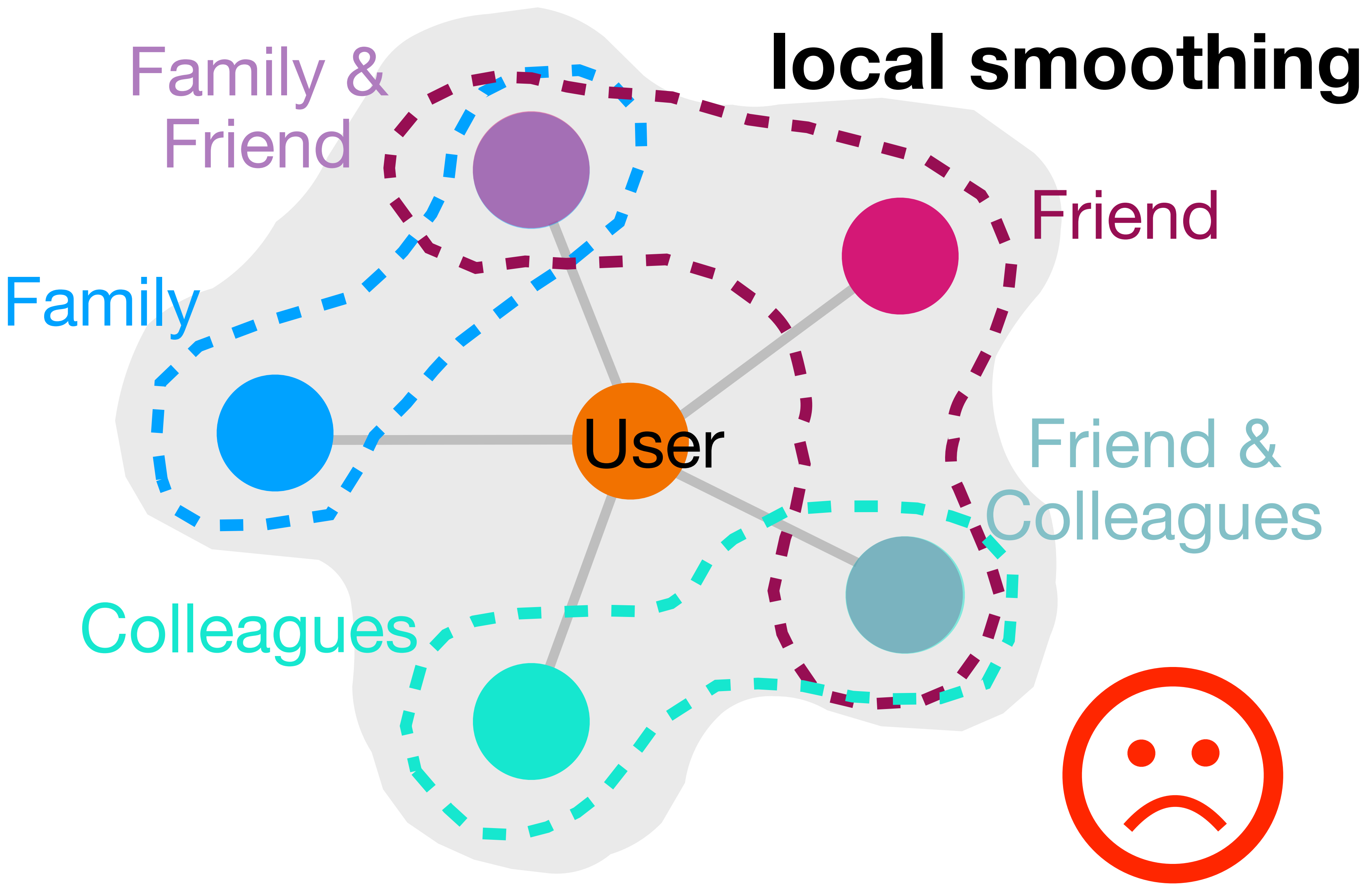
Conventional GNNs assume <u>homogenous node interactions</u> and employ <u>neighborhood smoothing</u>.

Research Challenges

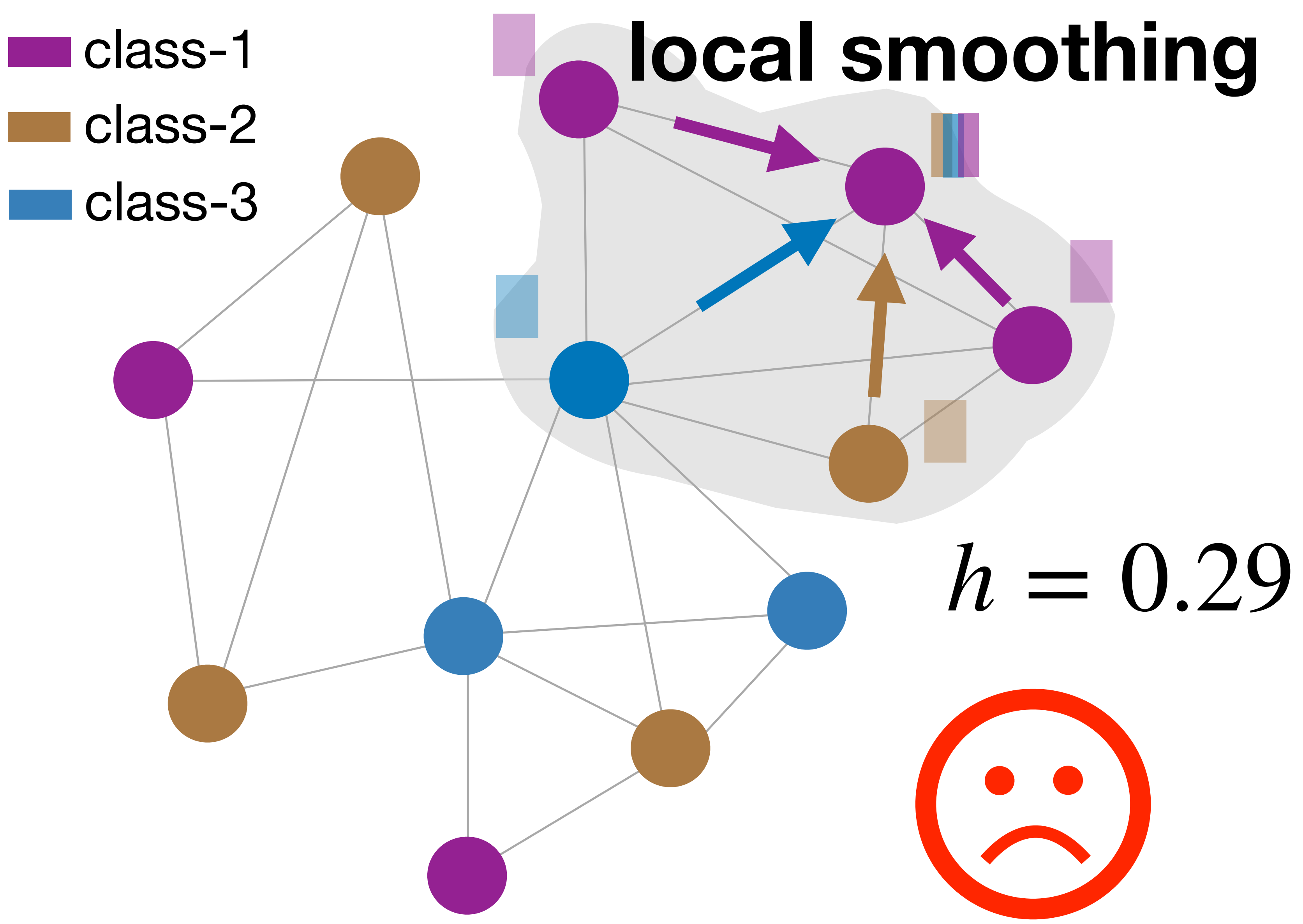
Real-world Scenarios

h = # intra-class edges / # total edges

- **Entangled Node Relationships**



- **Heterophilic Linking Patterns**



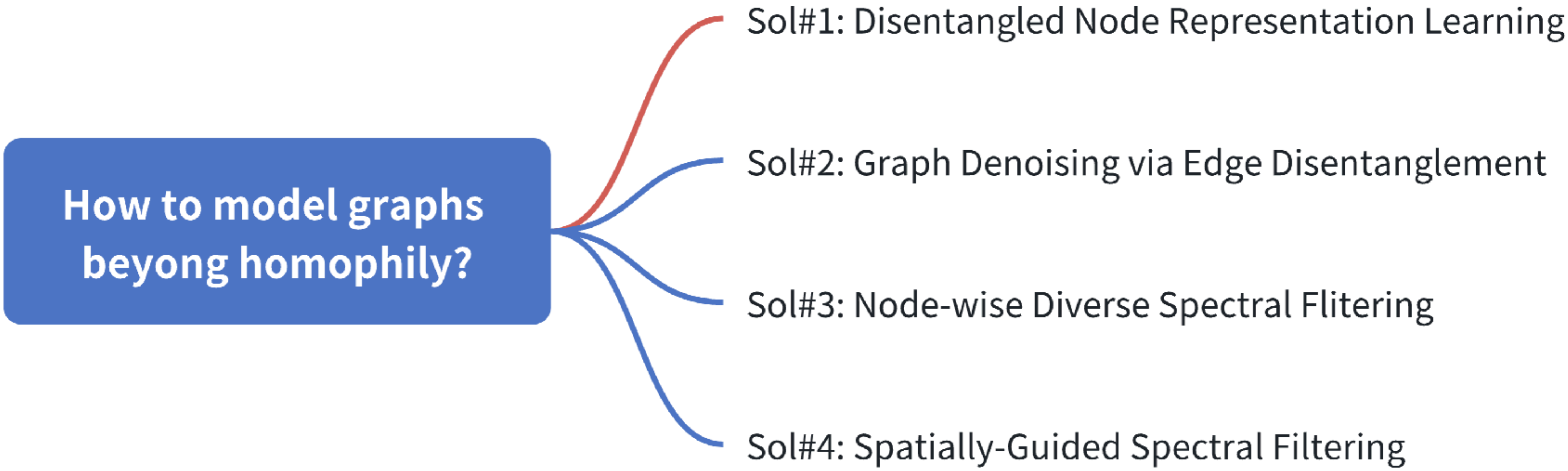
Conventional GNNs assume <u>homogenous node interactions</u> and employ <u>neighborhood smoothing</u>.

Proposed Solutions x 4

How to model graphs beyong homophily?

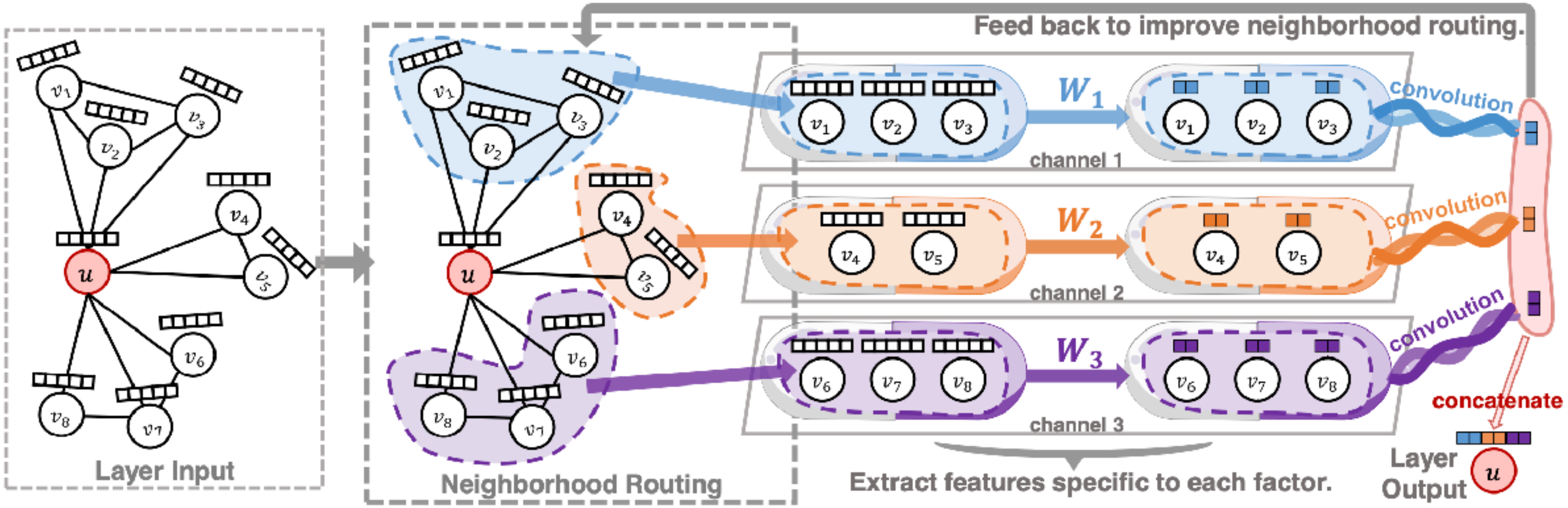
- Sol#1: Disentangled Node Representation Learning
- Sol#2: Graph Denoising via Edge Disentanglement
- Sol#3: Node-wise Diverse Spectral Flitering
- Sol#4: Spatially-Guided Spectral Filtering

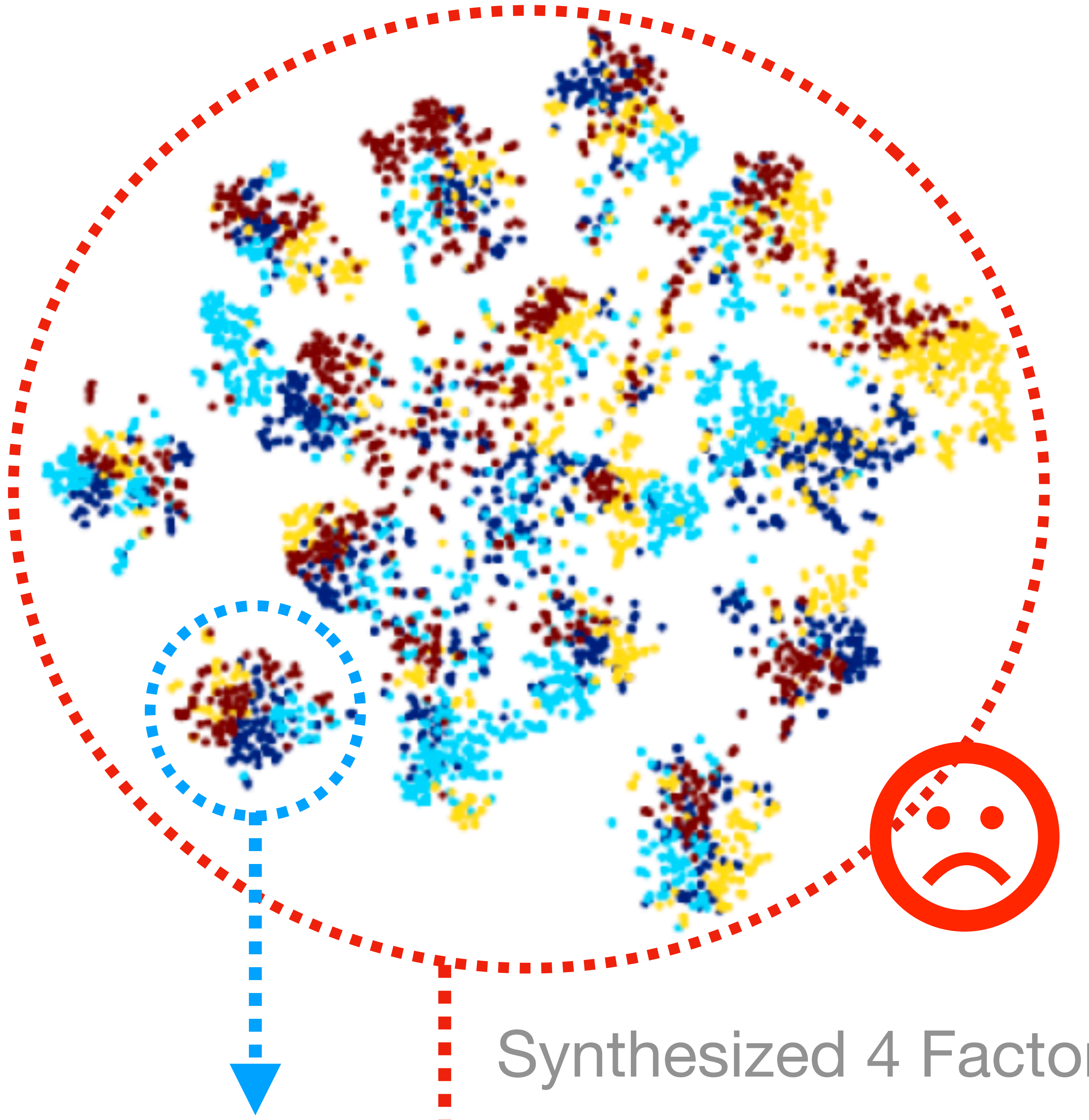
Solution #1: Disentangled Node Representation Learning



Solution #1: Disentangled Node Representation Learning

Our Findings (on previous works)





Synthesized 4 Factors

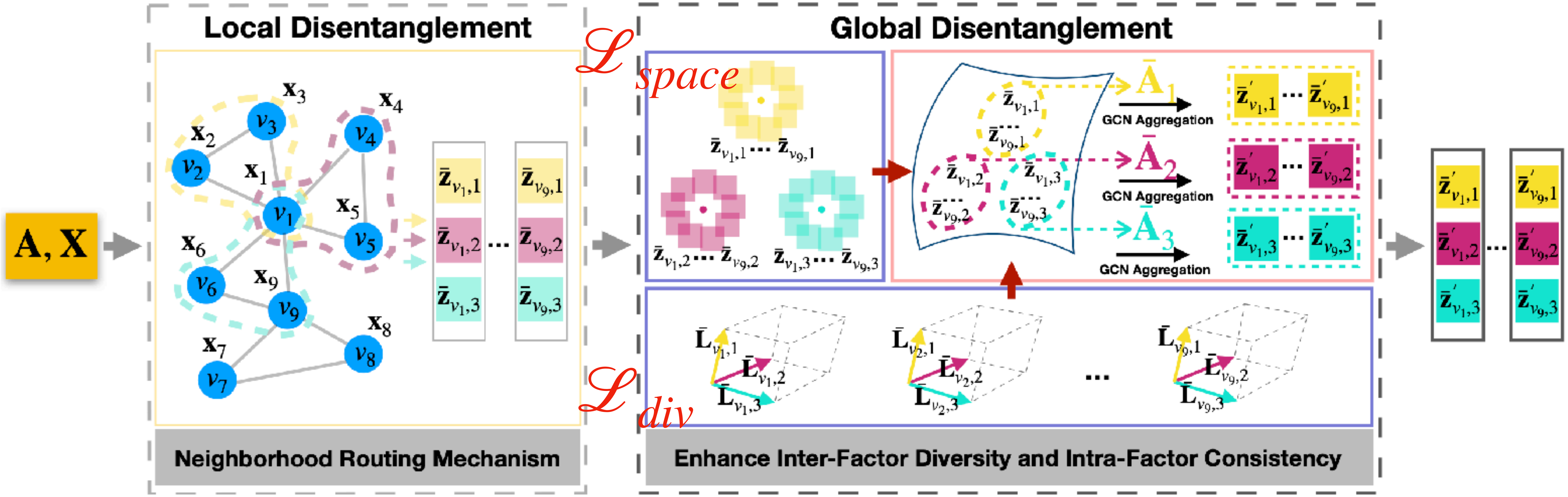
DisenGCN relies on a routing mechanism for local neighborhood partition.

"disentangled locally yet entangled globally"

Jianxin Ma, et.al., Disentangled Graph Convolutional Networks. *ICML, 2021*.

Solution #1: Disentangled Node Representation Learning

Our Modifications

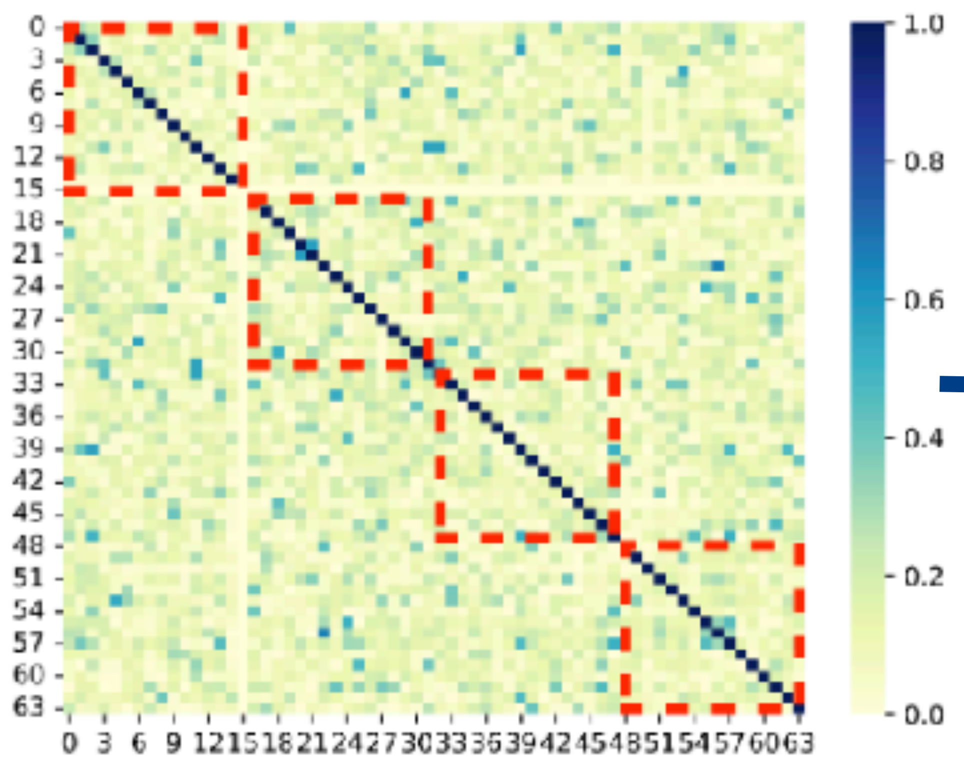


Our LGD-GCN enables global message passing over latent node relations to learn disentangled representation from complete graph views.

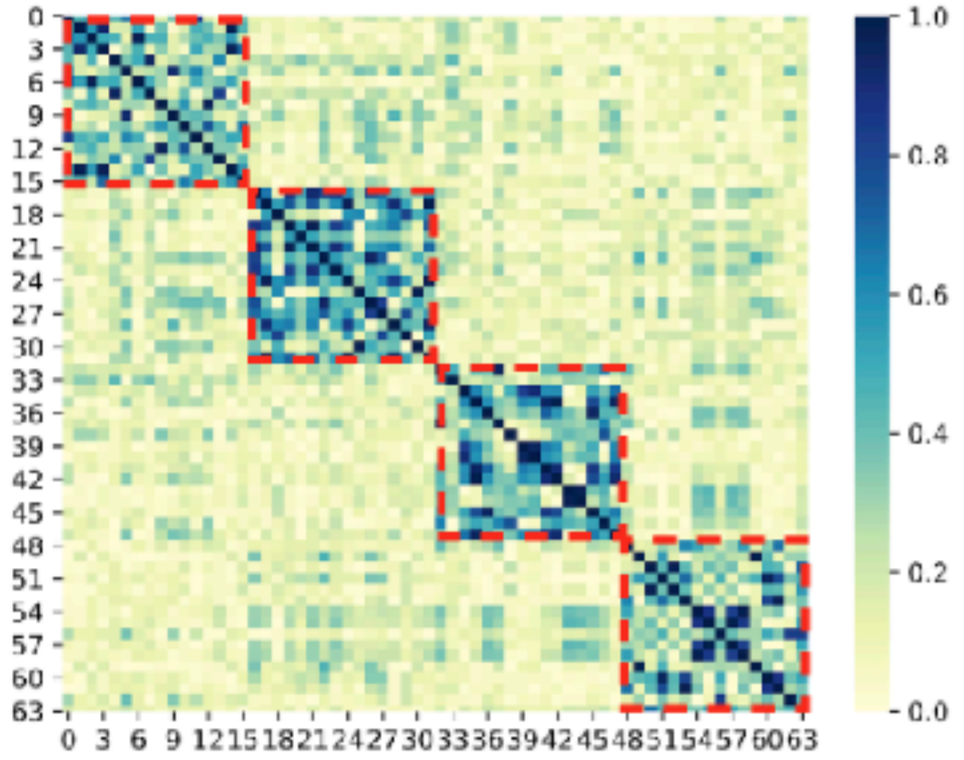
Jingwei Guo, et.al. Learning Disentangled Graph Convolutional Networks Locally and Globally. *TNNLS, 2022*.

Solution #1: Disentangled Node Representation Learning

Empirical Results



DisenGCN

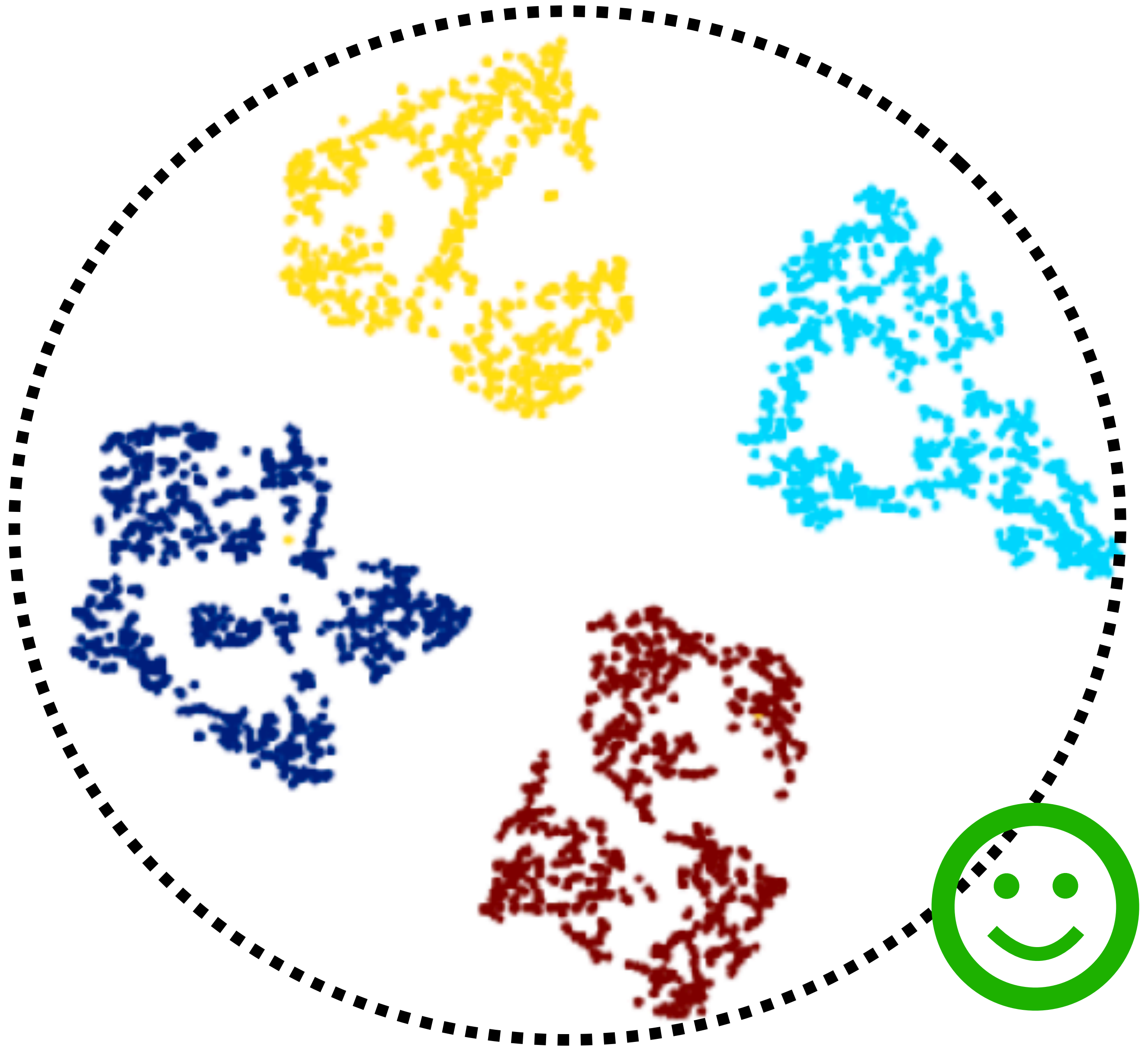


LGD-GCN (Ours)

Notable Block-wise Feature Correlation

Accuracy +7.4% on Social Nets

Methods	Blogcatalog		Flickr	
MoNet [69]	74.7±0.4	74.6±0.5	61.7±0.7	61.6±1.0
GCN [6]	73.8±0.3	72.9±0.4	56.6±0.4	57.6±0.3
GraphSAGE [10]	73.7±0.3	73.0±0.4	56.3±0.4	57.0±0.4
GAT [11]	56.7±5.0	57.5±3.2	45.1±1.0	45.1±1.4
SGC [12]	74.5±0.3	73.7±0.4	61.4±0.2	60.6±0.3
JK-Net [13]	76.5±0.3	75.8±0.5	64.6±0.4	64.1±0.4
DisenGCN [17]	86.5±1.3	86.4±1.2	75.8±0.6	76.7±0.6
IPGDN [18]	86.9±0.9	86.1±1.1	75.9±0.5	76.8±0.6
FactorGCN [47]	78.4±1.3	77.6±2.1	47.0±1.7	45.4±2.0
LGD-GCN (ours)	**93.7±0.4**	**93.9±0.4**	**85.5±0.6**	**84.0±0.8**

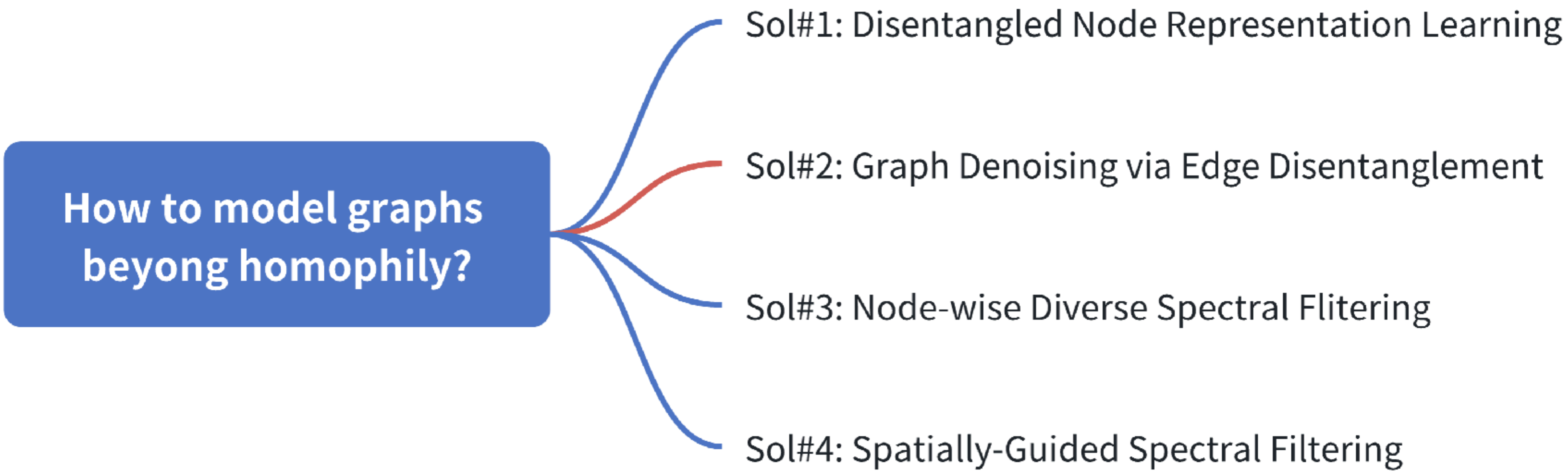


Synthesized 4 Factors

"disentangled *both locally and globally*"

Jingwei Guo, et.al. Learning Disentangled Graph Convolutional Networks Locally and Globally. *TNNLS, 2022*.

Solution #2: Graph Denoising via Edge Disentanglement



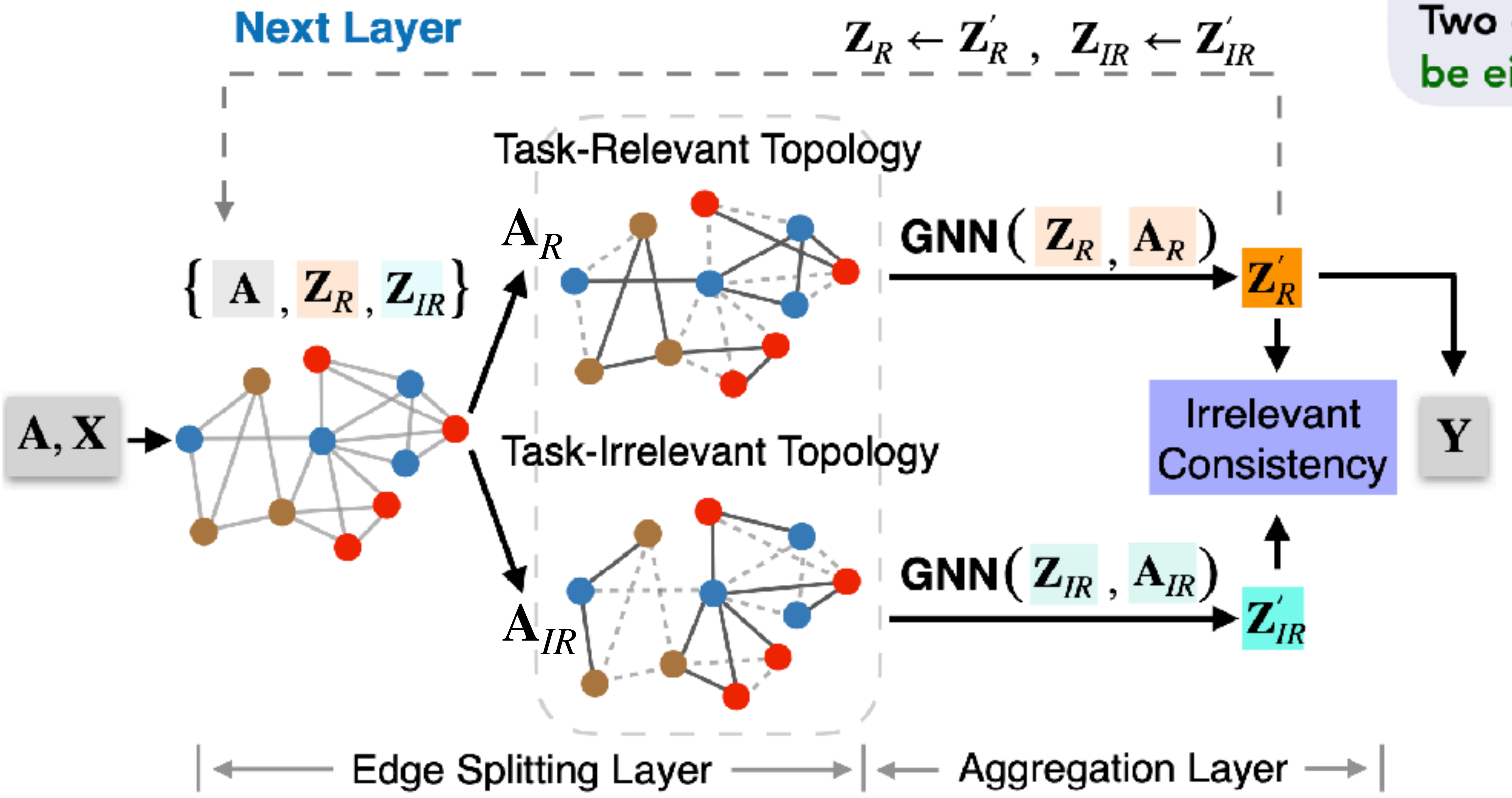
Solution #2: Graph Denoising via Edge Disentanglement

Conventional Smoothness Assumption

Two connected nodes mostly share task-beneficial similarity.

Disentangled Smoothness Assumption (Ours)

Two connected nodes share similarity in some features, which could be either relevant or irrelevant (even harmful) to learning tasks.



Our ES-GNN addresses heterophilic graphs by partitioning network topology, and disentangling node features.

Jingwei Guo, et.al. ES-GNN: Generalizing Graph Neural Networks Beyond Homophily with Edge Splitting. *TPAMI, 2024*.

Solution #2: Graph Denoising via Edge Disentanglement

Theoretical Justification

- **Conventional GNNs**

Graph Denoising Problem

$$\arg\min_{\mathbf{Z}} \ \|\mathbf{Z}-\mathbf{X}\|_2^2 + \xi \cdot tr(\mathbf{Z}^T\mathbf{L}\mathbf{Z})$$

Our Analysis $\downarrow$ $L = L_R + L_{IR}$

$$\arg\min_{\mathbf{Z}} \ \|\mathbf{Z}-\mathbf{X}\|_2^2 + \xi \cdot tr(\mathbf{Z}^T\mathbf{L}_R\mathbf{Z}) \boxed{+ \xi \cdot tr(\mathbf{Z}^T\mathbf{L}_{IR}\mathbf{Z})}$$

Possible classification-harmful information could be preserved in $\mathbf{Z}$

- **Our ES-GNN**

Disentangled Graph Denoising Problem

$$\arg\min_{\boxed{\mathbf{Z}_R,\mathbf{Z}_{IR}}} \ \|\mathbf{Z}_R-\mathbf{X}_{IR}\|_2^2 + \|\mathbf{Z}_{IR}-\mathbf{X}_{IR}\|_2^2 + \xi \cdot tr(\mathbf{Z}_R^T\mathbf{L}_R\mathbf{Z}_R) + \xi \cdot tr(\mathbf{Z}_{IR}^T\mathbf{L}_{IR}\mathbf{Z}_{IR})$$

$$\text{where} \quad \mathbf{L}_R = \mathbf{D}_R - \mathbf{A}_R,\ \mathbf{L}_{IR} = \mathbf{D}_{IR} - \mathbf{A}_{IR}$$

$$\text{s.t.} \quad \mathbf{A}_R + \mathbf{A}_{IR} = \mathbf{A}$$

$$\mathbf{A}_{R(i,j)}, \mathbf{A}_{IR(i,j)} \in [0,1].$$

Possible classification-harmful information can be excluded from $\mathbf{Z}_R$ & disentangled in $\mathbf{Z}_{IR}$

Jingwei Guo, et.al. ES-GNN: Generalizing Graph Neural Networks Beyond Homophily with Edge Splitting. *TPAMI, 2024*.

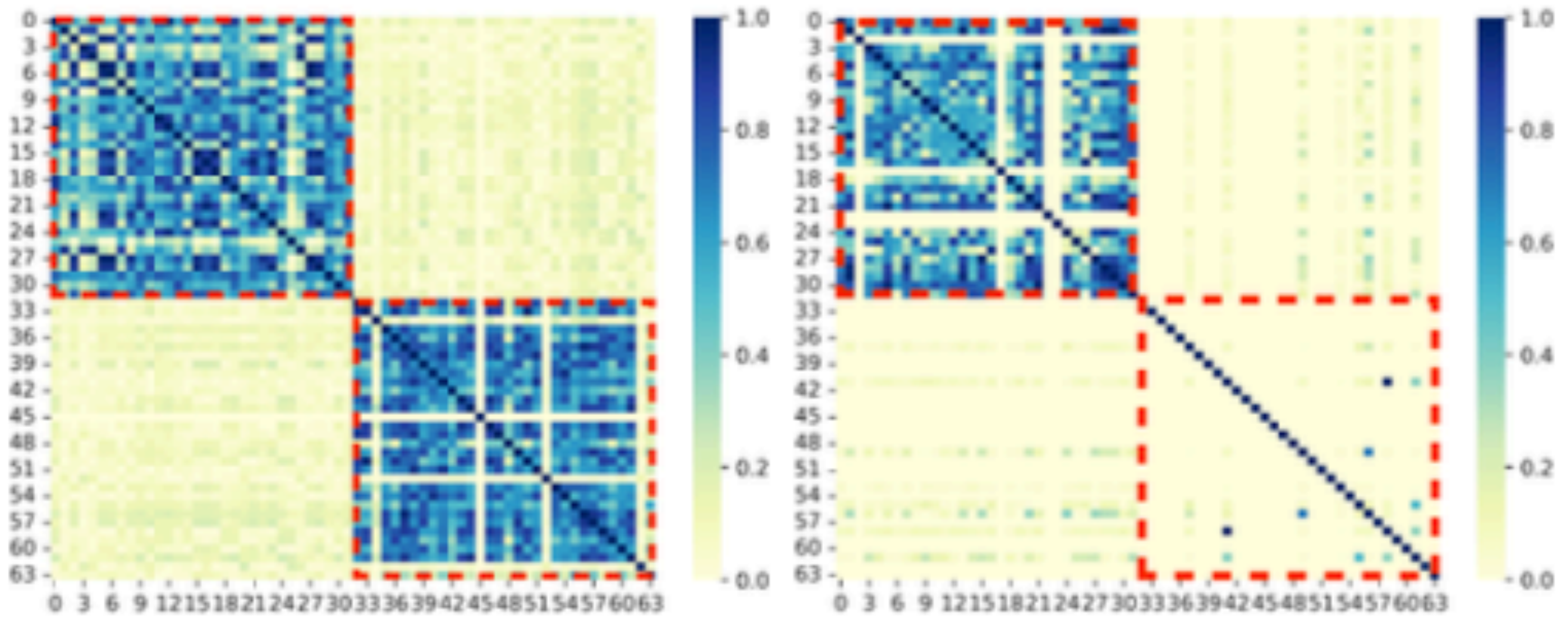
Solution #2: Graph Denoising via Edge Disentanglement

Empirical Results

Node classification accuracies (%) over 100 runs. Error Reduction gives the average improvement of ES-GNN upon baselines w/o Basic GNNs.

Datasets	Heterophilic Graphs							Homophilic Graphs			
	Squirrel	**Chameleon**	**Wisconsin**	**Cornell**	**Texas**	**Twitch-DE**	**Actor**	**Cora**	**Citeseer**	**Pubmed**	**Polblogs**
GCN [30]	55.2±1.5	67.6±2.0	59.5±3.6	52.8±6.0	61.7±3.7	74.0±1.2	31.2±1.3	79.7±1.2	69.5±1.7	78.7±1.6	89.4±0.9
SGC [6]	50.7±1.3	61.9±2.6	53.7±3.9	51.2±0.9	51.4±2.2	73.9±1.3	30.9±0.6	79.1±1.0	69.9±2.0	76.6±1.3	89.0±1.5
GAT [26]	54.8±2.2	67.3±2.2	57.9±4.5	50.4±5.9	55.4±5.9	73.7±1.3	30.5±1.2	82.0±1.1	69.9±1.7	78.6±2.0	87.4±1.1
NeuralSparse [48]	40.0±1.6	60.5±2.0	70.8±3.4	64.1±5.5	66.4±5.7	71.3±1.3	35.5±1.1	78.5±1.4	69.7±1.8	79.1±1.2	89.3±0.9
GCN-LPA [49]	54.2±1.1	63.4±1.9	63.3±3.7	65.6±7.3	61.2±7.6	74.0±1.2	37.8±0.9	80.4±1.5	69.7±1.7	79.7±1.3	**89.7±0.8**
DisenGCN [66]	42.4±1.6	58.4±2.3	78.1±4.0	77.4±4.4	71.3±5.7	73.5±1.7	36.7±1.2	81.5±1.3	69.2±1.7	80.0±1.6	89.5±0.9
FactorGCN [33]	56.6±2.4	69.8±2.0	64.2±4.8	50.6±1.8	69.5±6.5	73.1±1.4	29.0±1.4	75.2±1.6	61.6±2.0	72.9±2.3	87.9±1.7
VEPM [71]	50.3±1.7	67.3±2.1	55.6±4.9	51.2±7.0	55.8±4.3	73.3±1.2	29.3±1.1	82.2±1.2	69.1±1.9	78.8±1.6	89.5±0.9
DisGNN [72]	55.1±4.8	68.2±1.9	54.6±5.4	52.0±5.7	60.6±3.9	69.2±0.8	30.2±1.3	78.2±1.4	66.2±2.2	77.6±1.7	89.6±0.9
GEN [13]	36.0±4.0	57.6±3.1	83.3±3.6	81.0±3.9	78.3±8.0	74.1±1.4	37.3±1.4	79.8±1.3	69.7±1.6	78.9±1.7	89.6±1.4
WRGAT [14]	39.6±1.4	57.7±1.6	82.9±4.5	79.2±3.5	80.5±6.1	70.0±1.3	38.6±1.1	71.7±1.5	64.1±1.9	73.3±2.1	88.2±1.2
H2GCN [10]	45.1±1.9	62.9±1.9	82.6±4.0	79.6±4.9	79.8±7.3	73.1±1.5	38.4±1.0	81.4±1.4	68.7±2.0	78.0±2.0	89.0±1.0
FAGCN [18]	50.4±2.6	68.9±1.8	82.3±4.4	79.4±5.5	80.3±5.5	74.1±1.4	37.9±1.0	82.6±1.3	70.3±1.6	80.0±1.7	89.3±1.1
GPR-GNN [11]	54.1±1.6	69.6±1.7	82.7±4.1	79.9±5.3	81.7±4.9	74.0±1.6	38.0±1.1	81.5±1.5	69.6±1.7	79.8±1.3	89.5±0.8
GloGNN++ [23]	63.3±1.2	71.4±2.0	84.9±4.2	82.0±3.5	81.4±5.6	72.8±1.1	38.2±1.2	80.9±1.4	70.5±1.9	76.8±2.1	89.6±0.8
ACM-GCN [46]	**67.0±1.3**	**75.3±2.2**	84.3±4.5	82.1±4.9	82.2±5.9	74.2±0.9	36.6±1.0	81.3±1.0	69.4±1.7	79.5±1.4	89.6±0.9
GOAL [47]	57.9±0.9	71.3±2.0	70.5±5.1	54.9±6.6	72.0±7.4	68.5±1.5	36.3±1.0	80.6±1.4	69.7±2.0	78.7±1.3	88.7±1.6
ES-GNN (ours)	62.4±1.4	72.3±2.1	**85.3±4.6**	**82.2±4.0**	**82.3±5.7**	**74.7±1.1**	**38.9±0.8**	**83.0±1.1**	**70.7±1.7**	**80.7±1.4**	**89.7±0.9**
Error Reduction	**11.5%**	**6.4%**	**11.0%**	**11.7%**	**9.4%**	**2.2%**	**3.2%**	**3.3%**	**2.3%**	**2.6%**	**0.5%**

Heterophilic vs. Homophilic



(a) Chameleon (b) Cora

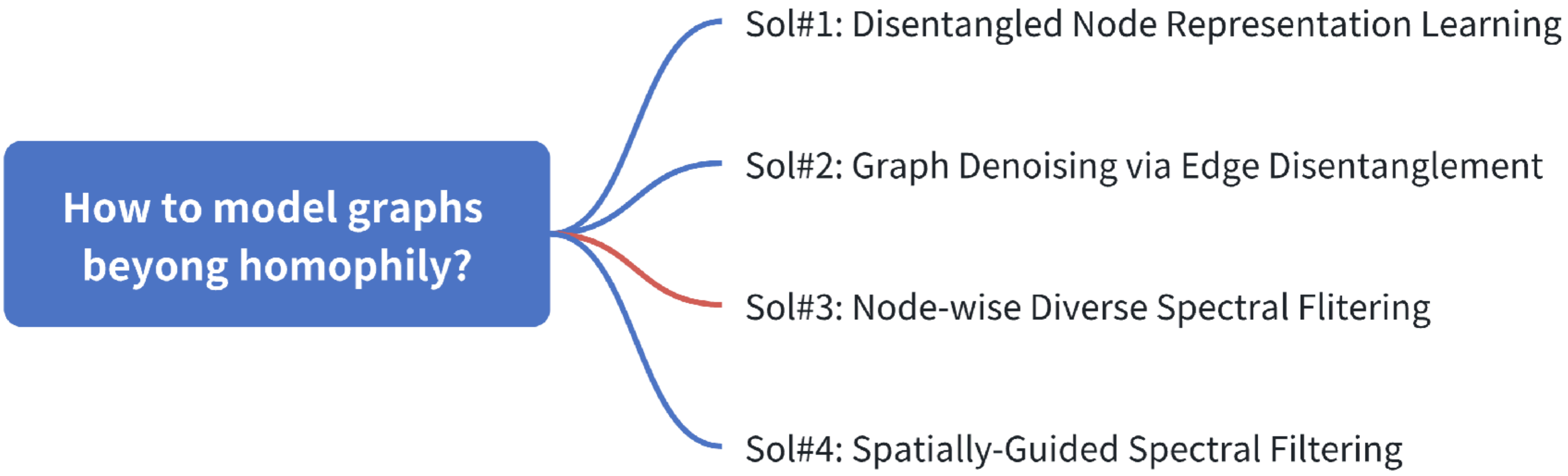
Block #1: Task-Relevant

Block #2: Task-Irrelevant

ES-GNN disentangle task-relevant and irrelevant features, showcasing universal adaptivity on different network types.

Jingwei Guo, et.al. ES-GNN: Generalizing Graph Neural Networks Beyond Homophily with Edge Splitting. *TPAMI, 2024*.

Solution #3: Node-wise Diverse Spectral Filtering

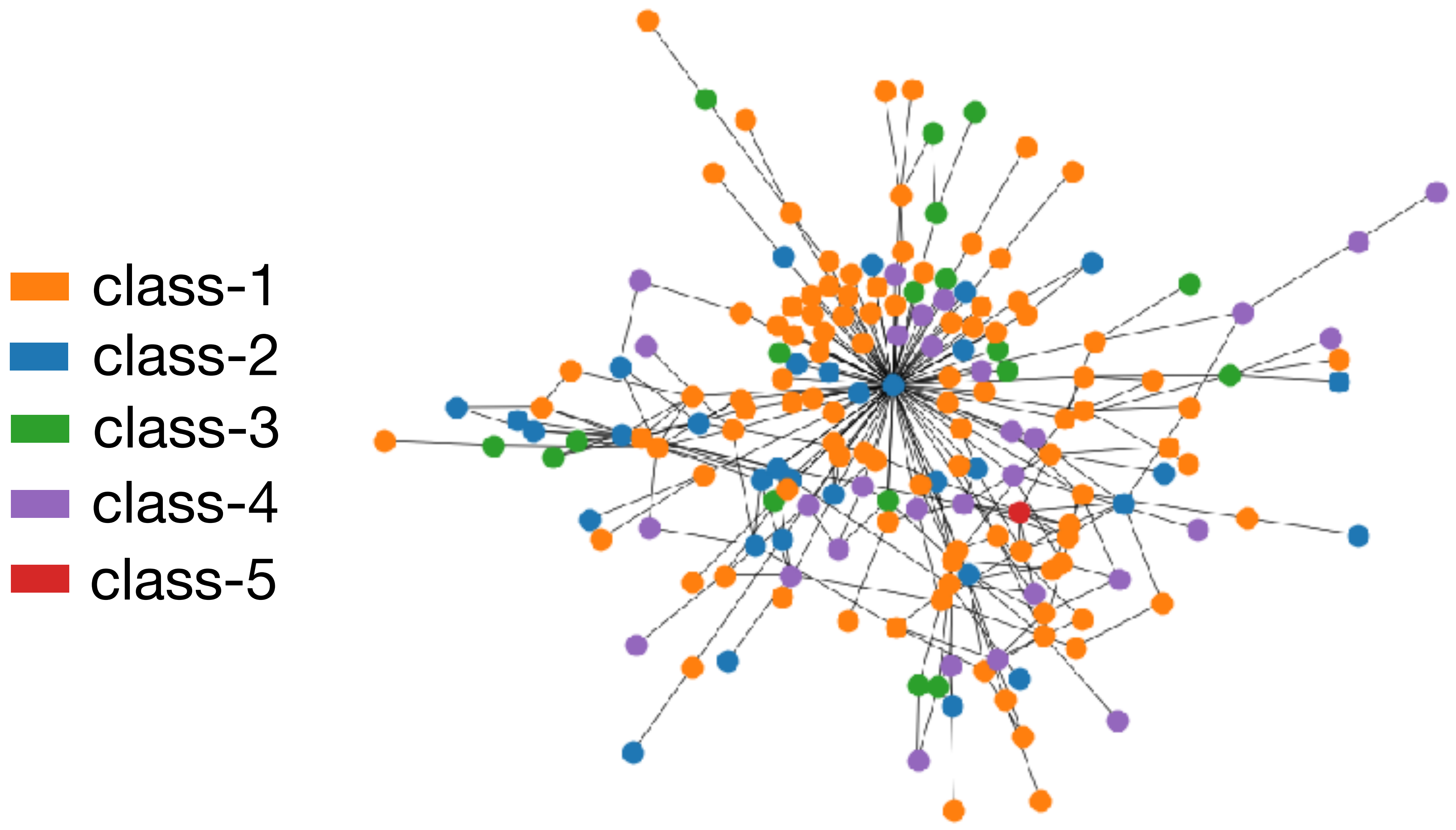


Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

Solution #3: Node-wise Diverse Spectral Filtering

New Heterophily Insights From a Broader Graph Context

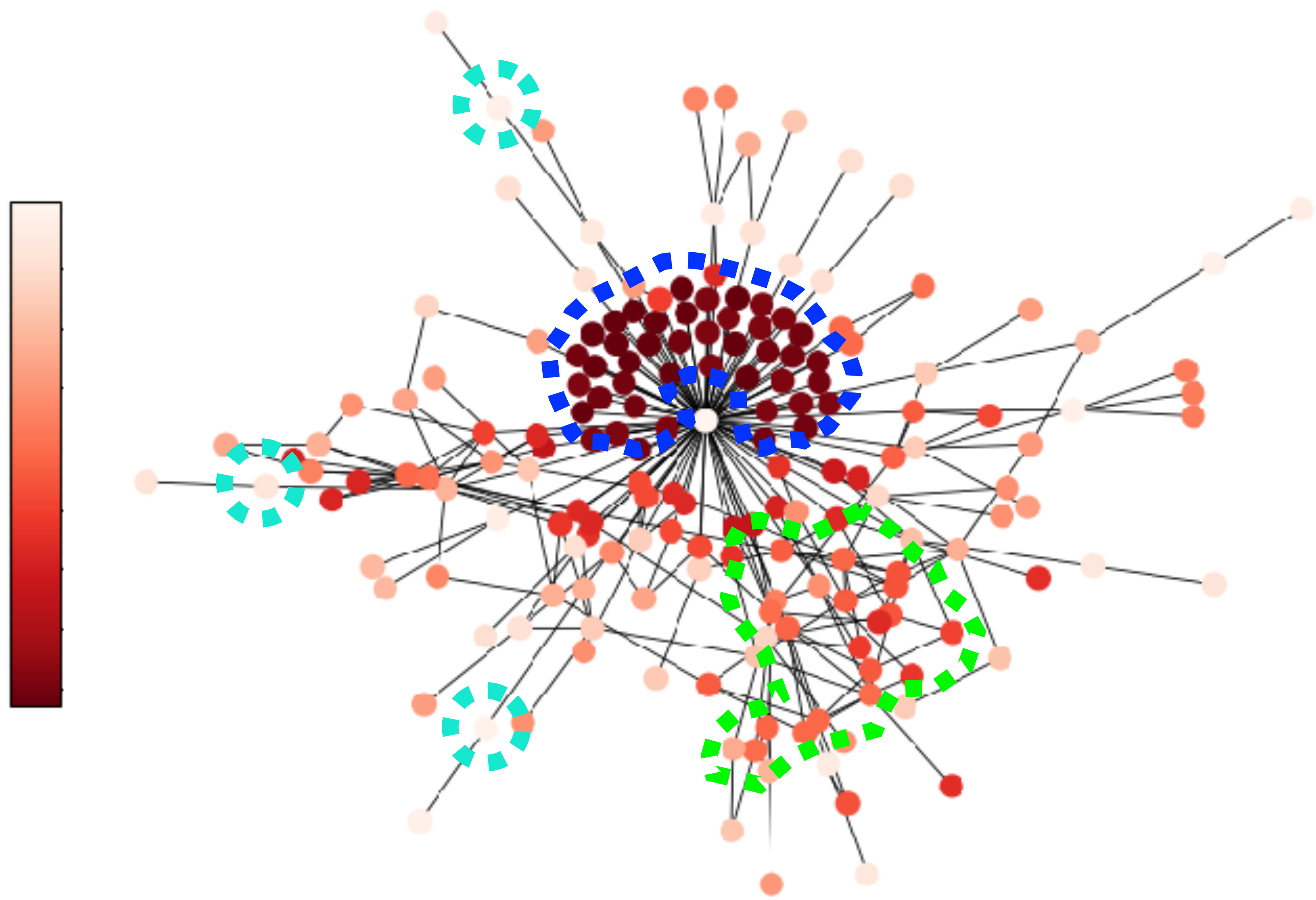
- Edge-level



label similarity / dissimilarity

Pairwise Distinction

- Subgraph-level



structure similarity / dissimilarity

Regional Disparity

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023.*

Solution #3: Node-wise Diverse Spectral Filtering

$$\mathbf{Z} = g_\psi(\hat{\mathbf{L}})\mathbf{X} = \sum_{k=0}^{K} \psi_k P_k(\hat{\mathbf{L}})\mathbf{X}$$

Filter Function

Polynomial Approx. with Shared Parameters for each K-hop Neighborhood

Most spectral GNNs assumes homogenous distributions between different graph regions. ☹

$$\mathbf{Z} = \sum_{k=0}^{K} \begin{pmatrix} \beta_{k,1} & 0 & \cdots & 0 \\ 0 & \beta_{k,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \beta_{k,N} \end{pmatrix} P_k(\hat{\mathbf{L}})\mathbf{X}$$

We augment the original parameters into node-specific filter weights to model diverse regional patterns. ☺

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

Solution #3: Node-wise Diverse Spectral Filtering

Definition 1 (Local Label Homophily). We define the Local Label Homophily as a measure of the local homophily level surrounding each node v_i:

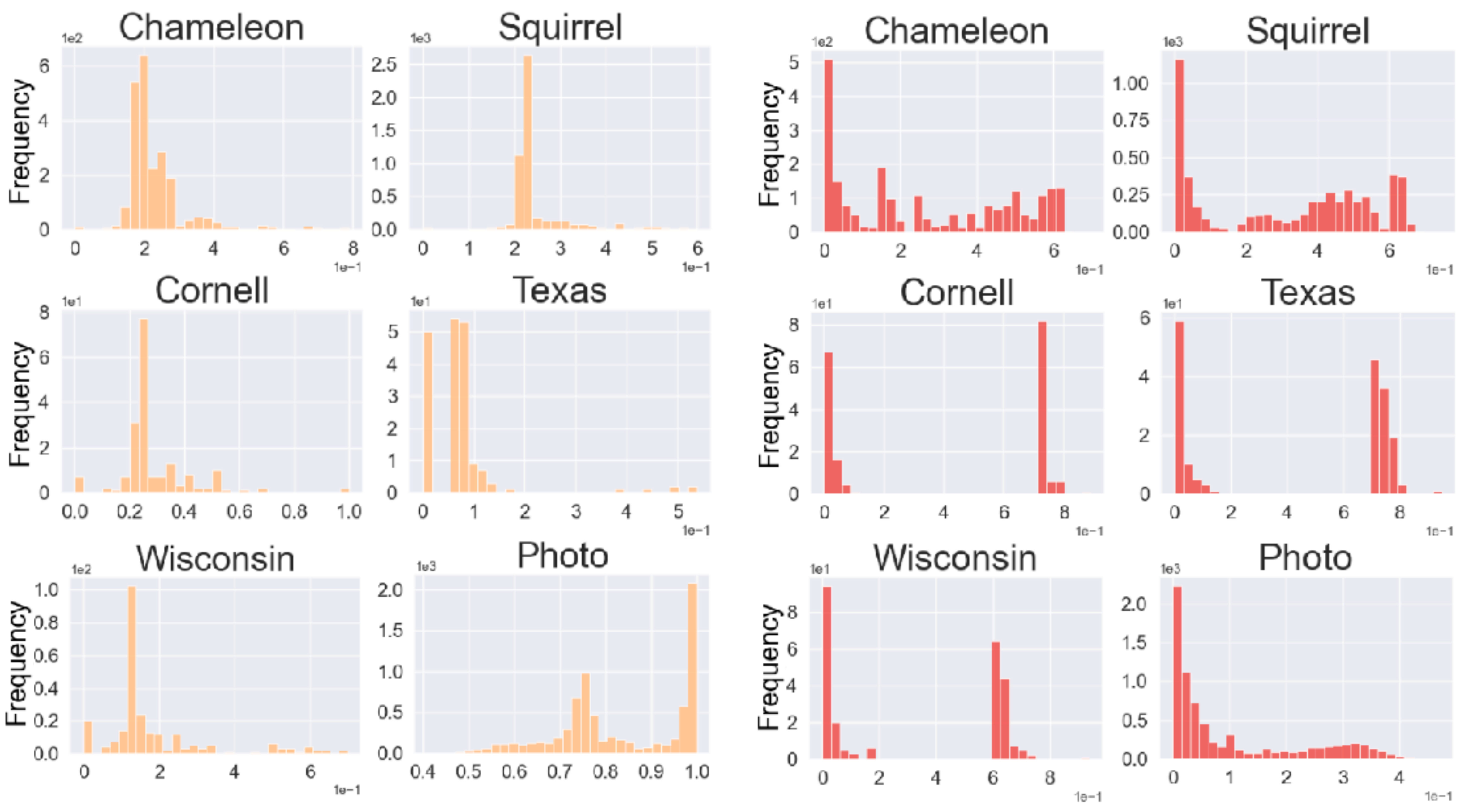
$$h_i = \frac{|\{(v_p, v_q)|\mathbf{y}_p = \mathbf{y}_q \wedge (v_p, v_q) \in \mathcal{E}_{i,k}\}|}{|\mathcal{E}_{i,k}|}$$

Here, h_i directly computes the edge homophily ratio [50] on the subgraph made up of the k-hop neighbors, and $\mathcal{E}_{i,k} = \{(v_p, v_q)|v_p, v_q \in \mathcal{N}_{i,k} \wedge (v_p, v_q) \in \mathcal{E}\}$ denotes its edge set.

Definition 2 (Local Graph Frequency). The Local Graph Frequency is defined by measuring the local smoothness level of the decomposed Laplacian eigenbases, and for each node v_i we have:

$$\lambda_{n,i} = \sum_{(v_p,v_q)\in\mathcal{E}_{i,k}} \left(\frac{1}{\sqrt{\deg_p}}\mathbf{u}_{n,p} - \frac{1}{\sqrt{\deg_q}}\mathbf{u}_{n,q}\right)^2$$

where $\lambda_{n,i}$ denotes the frequency or smoothness level of each Laplacian eigenbasis $\mathbf{u}_n$ upon the subgraph induced by the k-hop neighbors. Since all summed elements in Eq. 1 are positive and $\mathcal{E}_{i,k} \subseteq \mathcal{E}$, we can always have a $\xi_i \in (0, 1)$ such that $\lambda_{n,i} = \xi_i \lambda_n$.



(a) Local Label Homophily **(b) Local Graph Frequency**

Regional disparity can be observed on real-world graphs via local metrics.

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023.*

Solution #3: Node-wise Diverse Spectral Filtering

Reshaping Homogenous Spectral Filtering

Scalar coefficient — Global graph frequency

$$\mathbf{Z} = \sum_{n=1}^{N} \tilde{S}_n \cdot \mathbf{U}_n \qquad \tilde{S}_n = \sum_{k=0}^{K} \alpha_k P_k(\lambda_n) \mathbf{U}_n^T \mathbf{X}$$

Hadamard product — The i-th element of vector coefficients

$$\mathbf{Z} = \sum_{n=1}^{N} \tilde{\mathbf{S}}_n \odot \mathbf{U}_n \qquad \tilde{\mathbf{S}}_n(i) = \sum_{k=0}^{K} \alpha_k P_k(\lambda_{n,i}) \mathbf{U}_n^T \mathbf{X}$$

Vector — Local graph frequency

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

$\mathbf{X}$ is taken as one-dimension as an example

Solution #3: Node-wise Diverse Spectral Filtering

Key Designs

- **Approximation Trick**

Proposition 1. Suppose a K-order polynomial function $f : [0,2] \to \mathbb{R}$ with polynomial basis $P_k(\cdot)$ and coefficients $\{\alpha_k\}_{k=0}^K$ in real number. For any pair of variables $x, \hat{x} \in [0,2]$ satisfying $x = \xi\hat{x}$ where ξ is a constant real number, we always have a function $g : [0,2] \to \mathbb{R}$ with the same polynomial basis but a different set of coefficients $\{\beta_k\}_{k=0}^K$ such that $f(x) = g(\hat{x})$.

$$\sum_{k=0}^{K} \mathbf{U}_n \alpha_k P_k(\lambda_{n,i}) \mathbf{U}_n^T \mathbf{X} = \sum_{k=0}^{K} \mathbf{U}_n \beta_{k,i} P_k(\lambda_i) \mathbf{U}_n^T \mathbf{X}$$

$$\mathbf{Z} = \sum_{k=0}^{K} \mathbf{diag}(\beta_{k,1}, \beta_{k,2}, \cdots, \beta_{k,N}) P_k(\hat{\mathbf{L}}) \mathbf{X}$$

- **Local & Global Decomposition**

global invariant graph properties

$$\beta_{k,i} \leftarrow \gamma_i \cdot \alpha_{k,i}$$

local diverse node contexts

- **Position-aware Filter Weights**

$$\arg\min_{\mathbf{P}} \|\mathbf{P}^{(0)} - \mathbf{P}\|_2^2 + \kappa_1 tr(\mathbf{P}^T \hat{\mathbf{L}} \mathbf{P}) + \kappa_2 \|\mathbf{P}^T\mathbf{P} - \mathbf{I}\|_2^2$$

$$\alpha_{k,i} = \sigma_p(\mathbf{W}^{(k)} \mathbf{P}_i^{(k)} + \mathbf{b}^{(k)}) \qquad k = 1,2,\cdots,K$$

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023.*

Solution #3: Node-wise Diverse Spectral Filtering

Empirical Results

Table 2: Node classification accuracies (%) ± 95% confidence interval over 100 runs.

Datasets	Heterophilic Graphs						Homophilic Graphs				
	Chameleon	Squirrel	Wisconsin	Cornell	Texas	Twitch-DE	Cora	Citeseer	Pubmed	Computers	Photo
GPR-GNN [9]	69.01±0.50	55.39±0.33	82.72±0.85	80.81±0.78	81.66±1.02	74.07±0.18	89.03±0.20	77.63±0.28	90.10±0.44	92.34±0.13	95.34±0.09
DSF-GPR-I	71.18±0.52	57.08±0.29	**87.64**±0.79	84.76±0.90	85.44±1.05	74.58±0.16	**89.64**±0.20	78.03±0.26	90.26±0.08	92.49±0.12	95.64±0.07
DSF-GPR-R	**71.64**±0.55	**58.44**±0.30	87.43±0.74	**84.93**±0.90	**85.56**±0.93	**74.81**±0.14	89.63±0.17	**78.22**±0.29	**90.51**±0.07	**92.80**±0.12	**95.73**±0.08
Our Improv.	2.63%	3.05%	4.92%	4.12%	3.9%	0.74%	0.61%	0.59%	0.41%	0.46%	0.39%
BernNet [20]	70.59±0.42	56.63±0.32	85.00±0.94	82.10±0.95	82.20±0.98	74.45±0.15	88.72±0.23	77.52±0.29	90.21±0.46	92.57±0.10	95.42±0.08
DSF-Bern-I	72.95±0.53	59.45±0.32	**88.23**±0.81	**85.07**±0.93	**84.59**±1.07	74.96±0.15	89.05±0.22	**78.32**±0.27	90.40±0.10	92.76±0.10	95.73±0.07
DSF-Bern-R	**73.60**±0.53	**59.99**±0.30	88.02±0.91	84.29±0.93	84.42±1.00	**75.00**±0.15	**89.10**±0.22	78.27±0.26	**90.52**±0.10	**92.84**±0.10	**95.79**±0.06
Our Improv.	3.01%	3.36%	3.23%	2.97%	2.39%	0.55%	0.38%	0.80%	0.31%	0.27%	0.37%
JacobiConv [42]	73.71±0.42	57.22±0.24	83.21±0.68	82.34±0.88	82.42±0.90	74.34±0.12	89.24±0.19	77.81±0.29	89.50±0.47	92.26±0.10	95.62±0.06
DSF-Jacobi-I	74.88±0.39	58.26±0.26	85.34±0.74	**84.54**±0.81	83.68±1.12	74.65±0.13	89.54±0.19	78.18±0.26	89.78±0.09	92.38±0.11	**95.76**±0.07
DSF-Jacobi-R	**75.00**±0.38	**59.23**±0.27	**86.13**±0.70	84.39±0.88	**84.46**±0.81	**74.75**±0.15	**89.66**±0.19	**78.23**±0.25	**90.07**±0.10	**92.44**±0.11	95.75±0.08
Our Improv.	1.29%	2.01%	2.92%	2.20%	2.04%	0.41%	0.42%	0.42%	0.41%	0.18%	0.14%

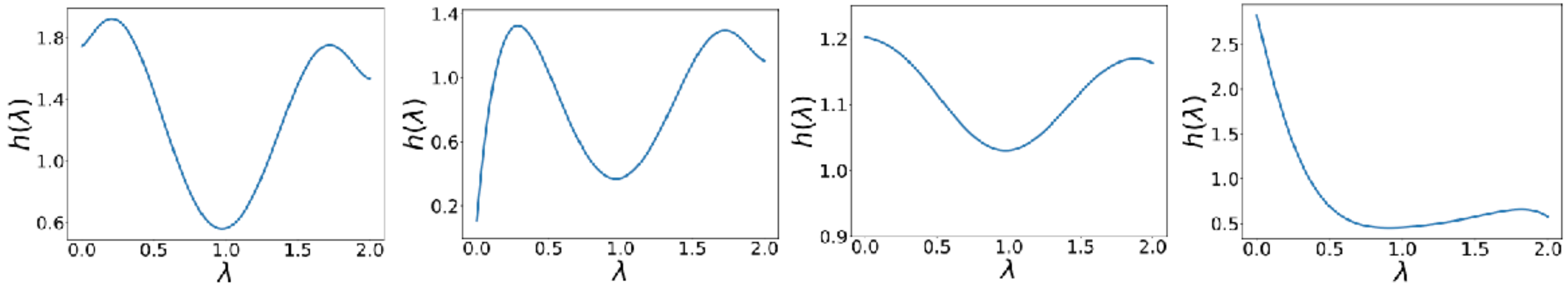
DSF can be readily plug-and-play in multiple spectral GNNs and consistently improve their performance.

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

Solution #3: Node-wise Diverse Spectral Filtering

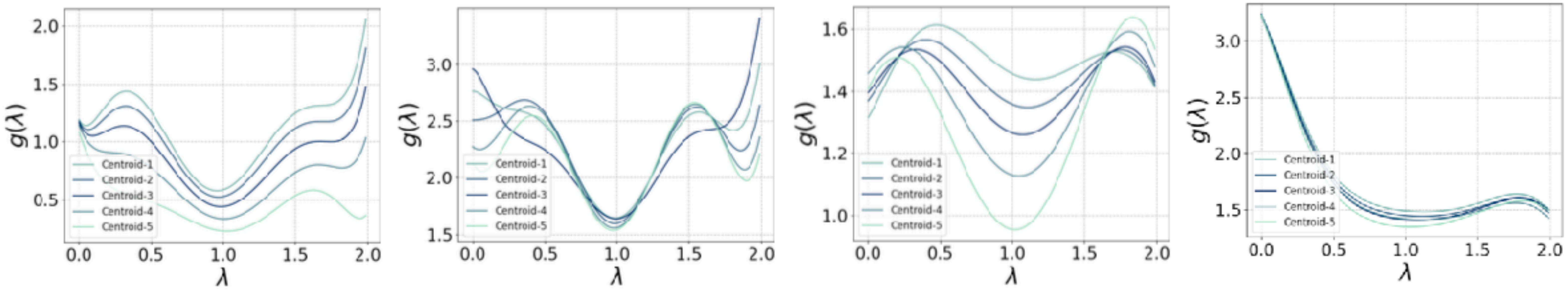
Empirical Results — Interpretable Analysis #1

- Traditional Spectral Filter (BernNet as a case):



(f) Chameleon (h) Squirrel (j) Cornell (b) CiteSeer

- Our Spectral Filter (DSF-Bern):

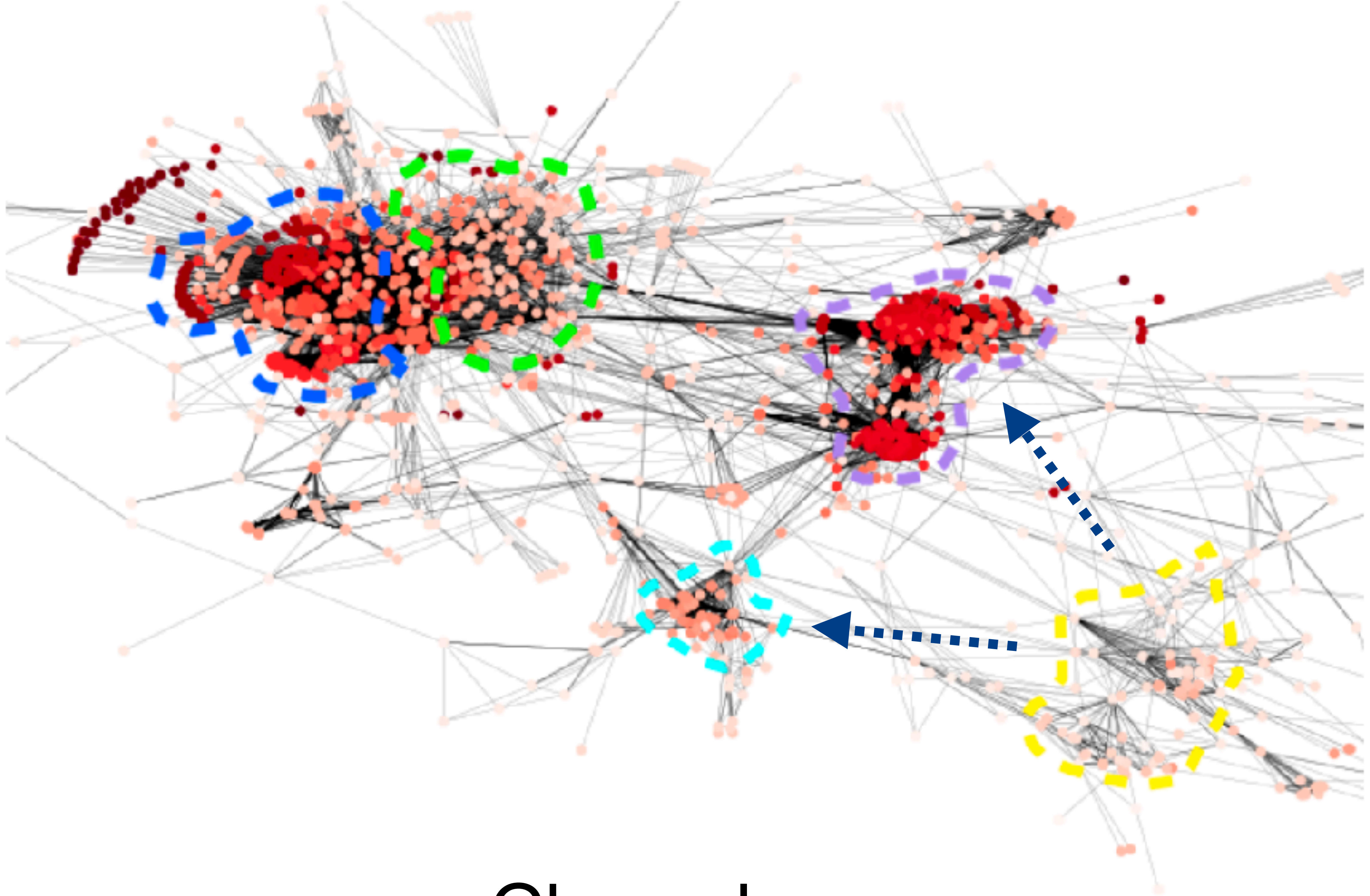


(a) Chameleon (b) Squirrel (c) Cornell (d) Citeseer

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

Solution #3: Node-wise Diverse Spectral Filtering

Empirical Results — Interpretable Analysis #2



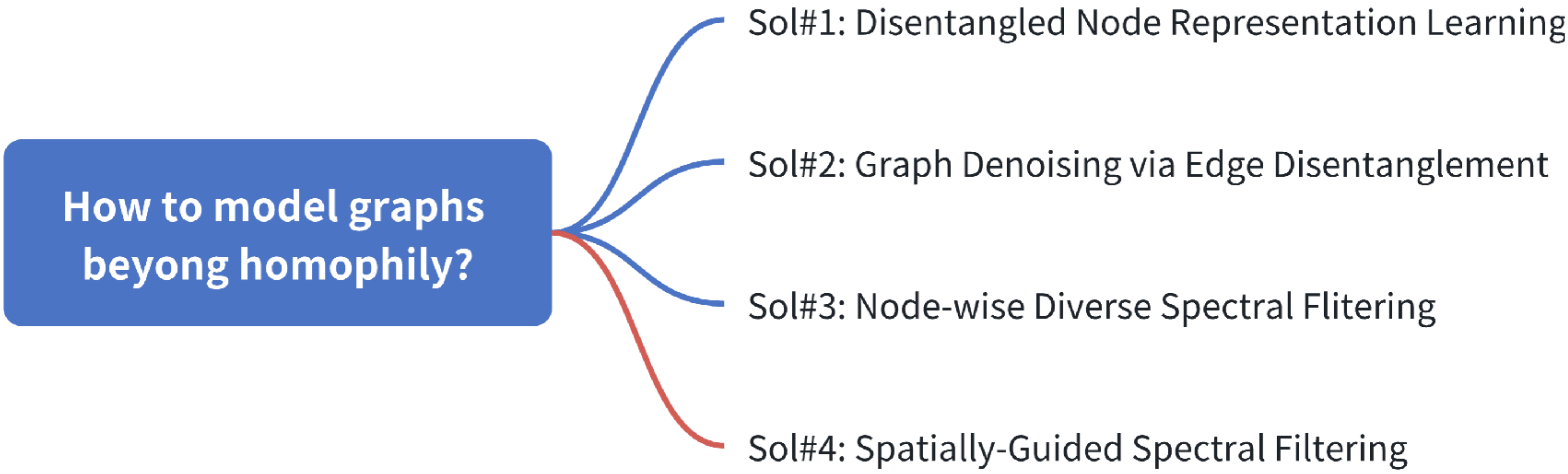
Chameleon

Squirrel

DSF captures regional disparity with node-specific filter weights.

Jingwei Guo, et.al. Graph Neural Networks with Diverse Spectral Filtering. *WWW, 2023*.

Solution #4: Spatially-Guided Spectral Filtering



Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Solution #4: Spatially-Guided Spectral Filtering

Deep Delve into Spectral GNNs

- $\mathbf{Z} = \mathbf{U} g_\psi(\mathbf{\Lambda}) \mathbf{U}^T \mathbf{X}$ theoretical spectral filtering

$$= g_\psi(\hat{\mathbf{L}})\mathbf{X} \approx \sum_{k=0}^{K} \psi_k P_k(\hat{\mathbf{L}})\mathbf{X} = \sum_{k=0}^{K} \omega_k \hat{\mathbf{A}}^k \mathbf{X}$$

practical spatial aggregation / message passing

- What information is essentially encoded by spectral GNNs in the spatial domain?

Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Solution #4: Spatially-Guided Spectral Filtering

Cross-Domain Interplay via a Generalized Graph Optimization

$$\arg\min_{\mathbf{Z}} \mathscr{L} = \alpha \|\mathbf{X} - \mathbf{Z}\|_2^2 + (1-\alpha)\mathrm{tr}(\mathbf{Z}^T \boxed{\gamma_\theta(\hat{\mathbf{L}})} \mathbf{Z})$$

Arbitrary Linking Patterns

- $\mathbf{Z}$ refer to node representations
- α is a trade-off coefficient
- $\gamma_\theta(\hat{\mathbf{L}}) = \mathbf{U}\gamma_\theta(\mathbf{\Lambda})\mathbf{U}^T$ determines propagation rate where $\gamma_\theta(\lambda) \geq 0$

Positive Semi-definite Constraint for Convexity Optimization

Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Solution #4: Spatially-Guided Spectral Filtering

- Closed-form Solution $\dfrac{\partial \mathcal{L}}{\partial \mathbf{Z}} = 0$

$$\mathbf{Z}^* = (\mathbf{I} + \frac{1-\alpha}{\alpha}\gamma_\theta(\hat{\mathbf{L}}))^{-1}\mathbf{X} = g_\psi(\hat{\mathbf{L}})\mathbf{X} = \mathbf{U}g_\psi(\mathbf{\Lambda})\mathbf{U}^T\mathbf{X}$$

Spectral Filter as a function of $\gamma_\theta(\cdot)$: $\boxed{g_\psi(\lambda) = (1 + \frac{1-\alpha}{\alpha}\gamma_\theta(\lambda))^{-1}}$

Spatial Aggregation

- Iterative Solution $\mathbf{Z}^{(k)} = \mathbf{Z}^{(k-1)} - \dfrac{1}{2}\dfrac{\partial \mathcal{L}}{\partial \mathbf{Z}}\Big|_{\mathbf{Z}=\mathbf{Z}^{(k-1)}}$

$$\mathbf{Z}^{(k)} = \alpha\mathbf{X} + (1-\alpha)\hat{\mathbf{A}}^{new}\mathbf{Z}^{(k-1)}$$

New Graph Structure: $\boxed{\hat{\mathbf{A}}^{new} = \mathbf{I} - \gamma_\theta(\hat{\mathbf{L}}) = \mathbf{I} - \frac{\alpha}{1-\alpha}(g_\psi(\hat{\mathbf{L}})^{-1} - \mathbf{I})}$

Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Solution #4: Spatially-Guided Spectral Filtering

Desirable Properties on $\hat{\mathbf{A}}^{new} = \mathbf{I} - \frac{\alpha}{1-\alpha}(g_\psi(\hat{\mathbf{L}})^{-1} - \mathbf{I})$

- Non-locality

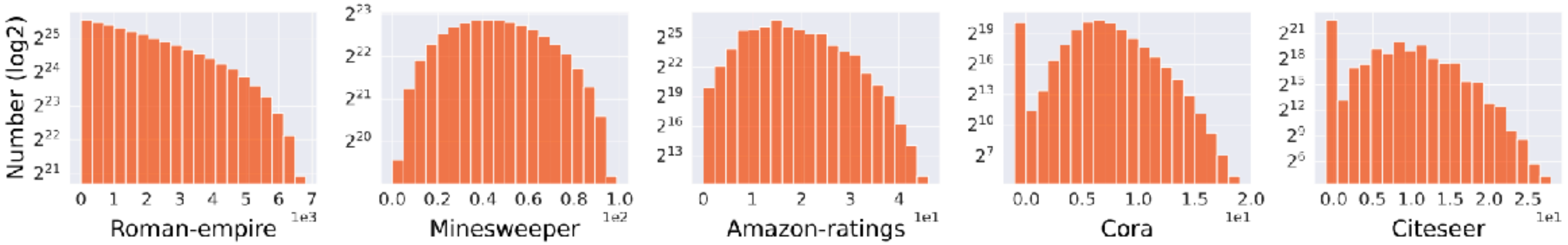


Figure 1: Distributions of connected nodes in the new graph based on their geodesic/shortest-path distance (as $\Delta_{i,j}$) in the original graph. Nodes, distant in the original graph ($\Delta_{i,j} > 1$ in x-axis), can be linked in the new graph (Number > 0 in y-axis).

- Signed edge weights reflecting "global node label relationships"

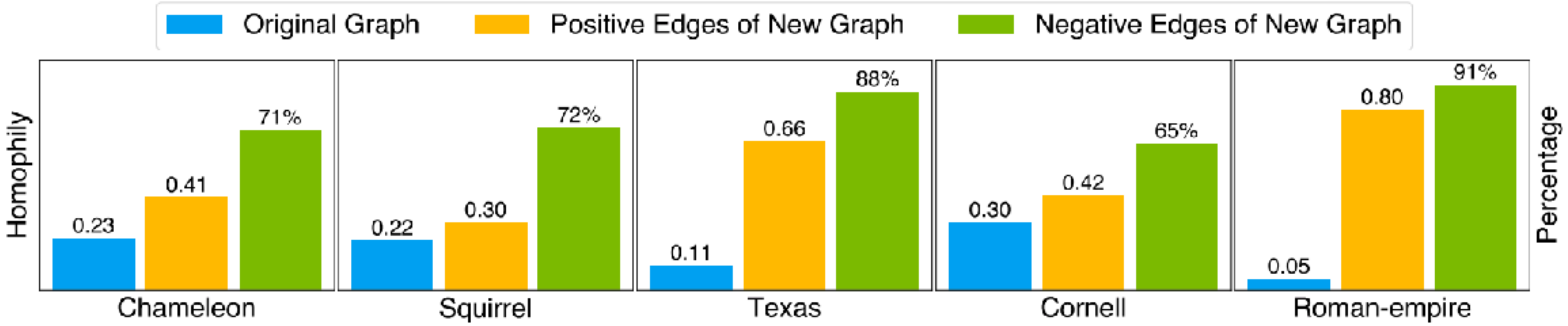


Figure 2: Left y-axis: Homophily comparison between original and new graphs, considering only positive edges (blue and yellow bars). Right y-axis: Percentage of edges connecting nodes from different classes, identified by negative edges (green bar).

Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Solution #4: Spatially-Guided Spectral Filtering

Overall Framework

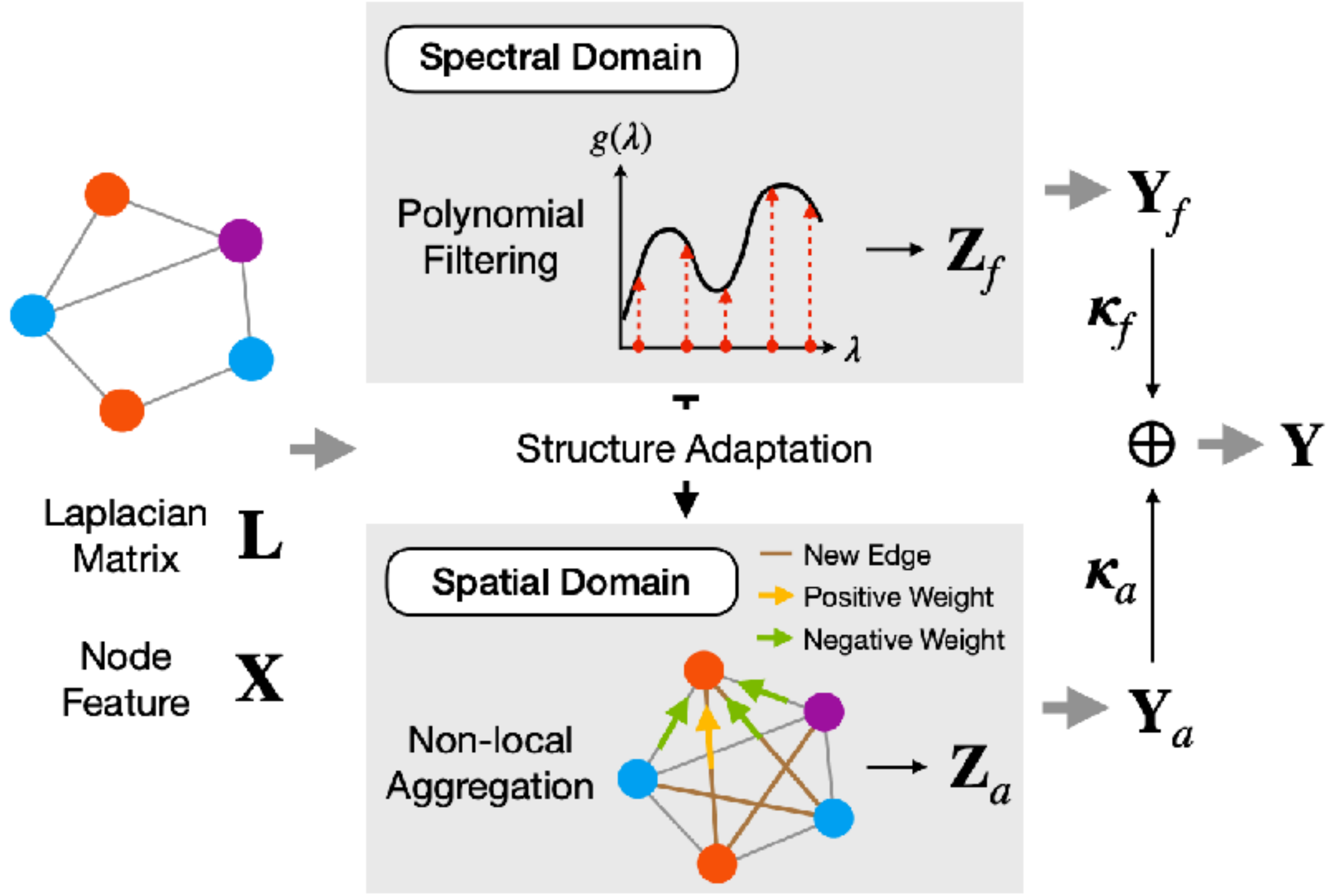


Figure 3: Illustration of the proposed SAF framework, where varying node colors represent different node labels.

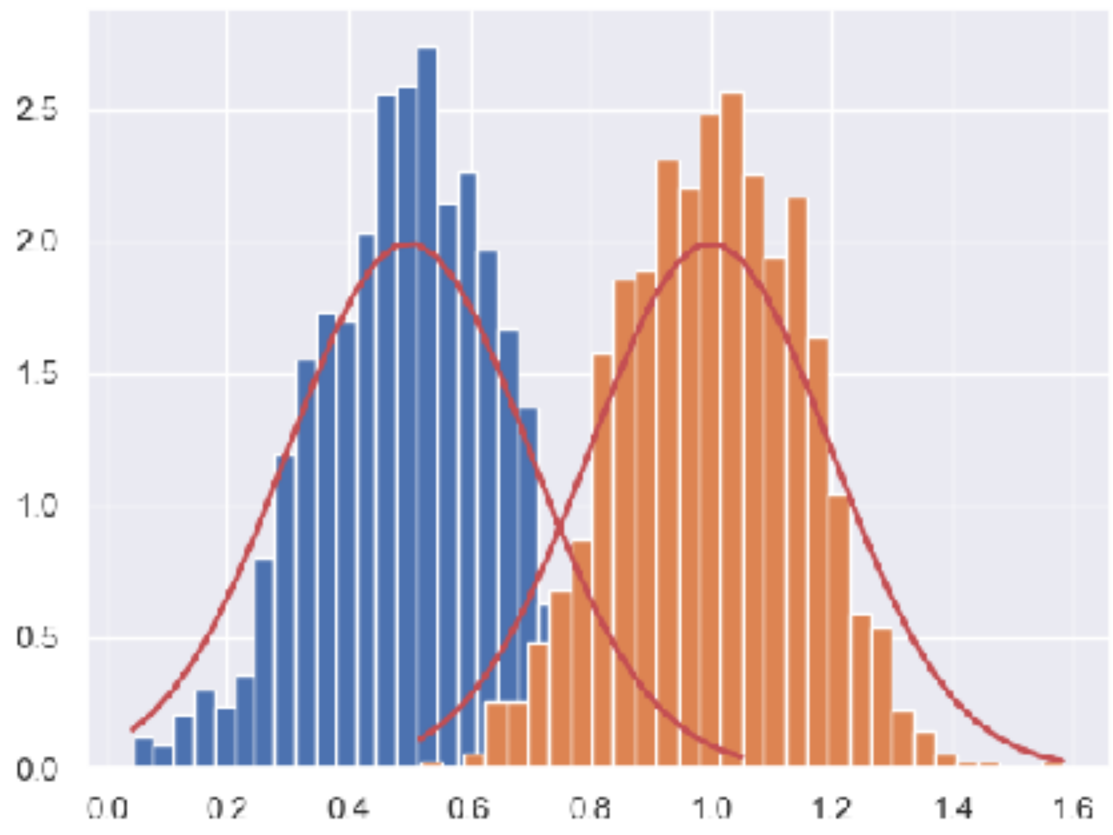
Our method leverages the adapted new graph by spectral filtering for non-local aggregation with signed edge weights.

- Capture long-range dependency
- Address heterophilic graph links

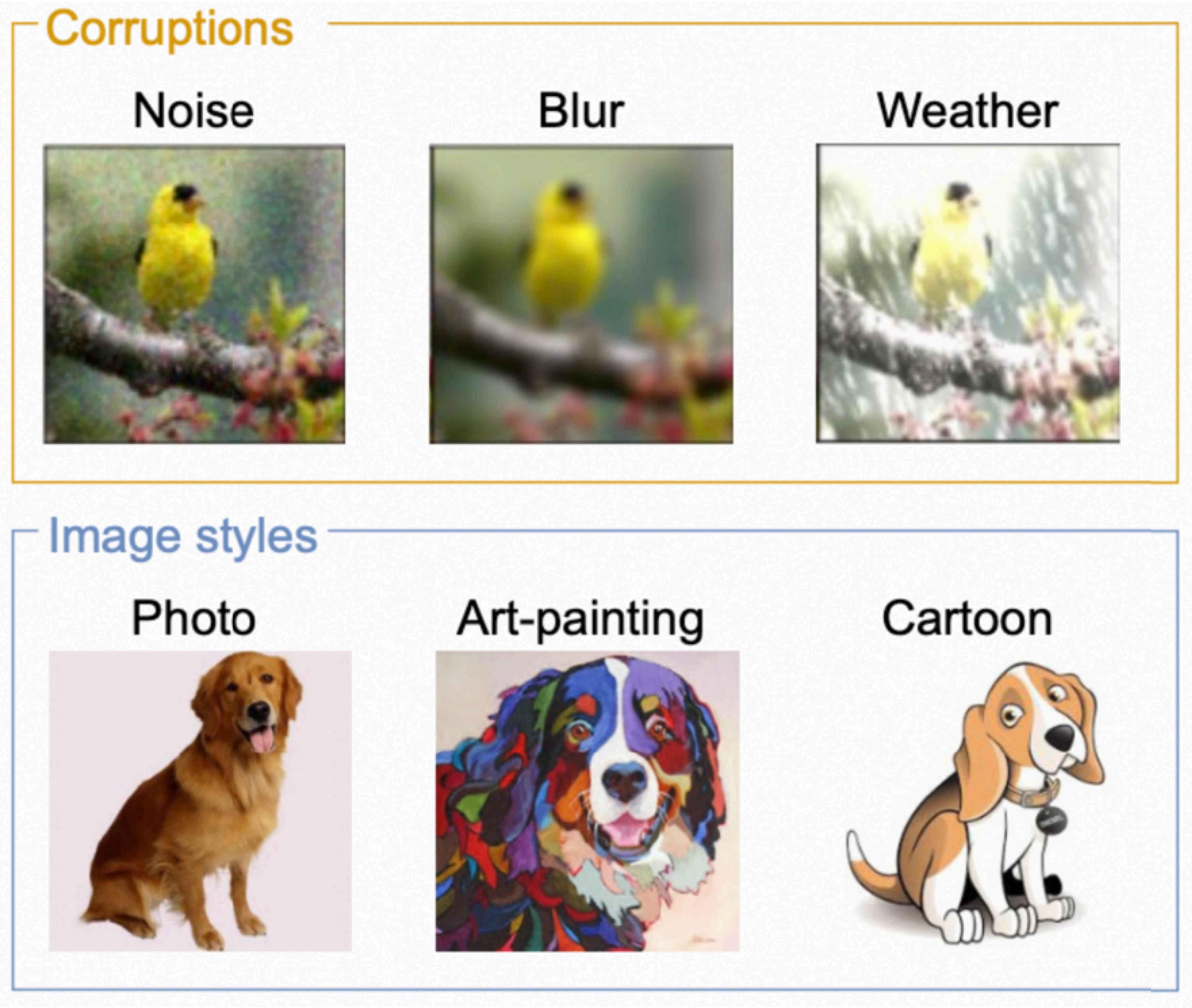
Jingwei Guo, et.al. Rethinking Spectral Graph Neural Networks with Spatially Adaptive Filtering. *TNNLS (Under 2nd Review), 2025*.

Research Project #2

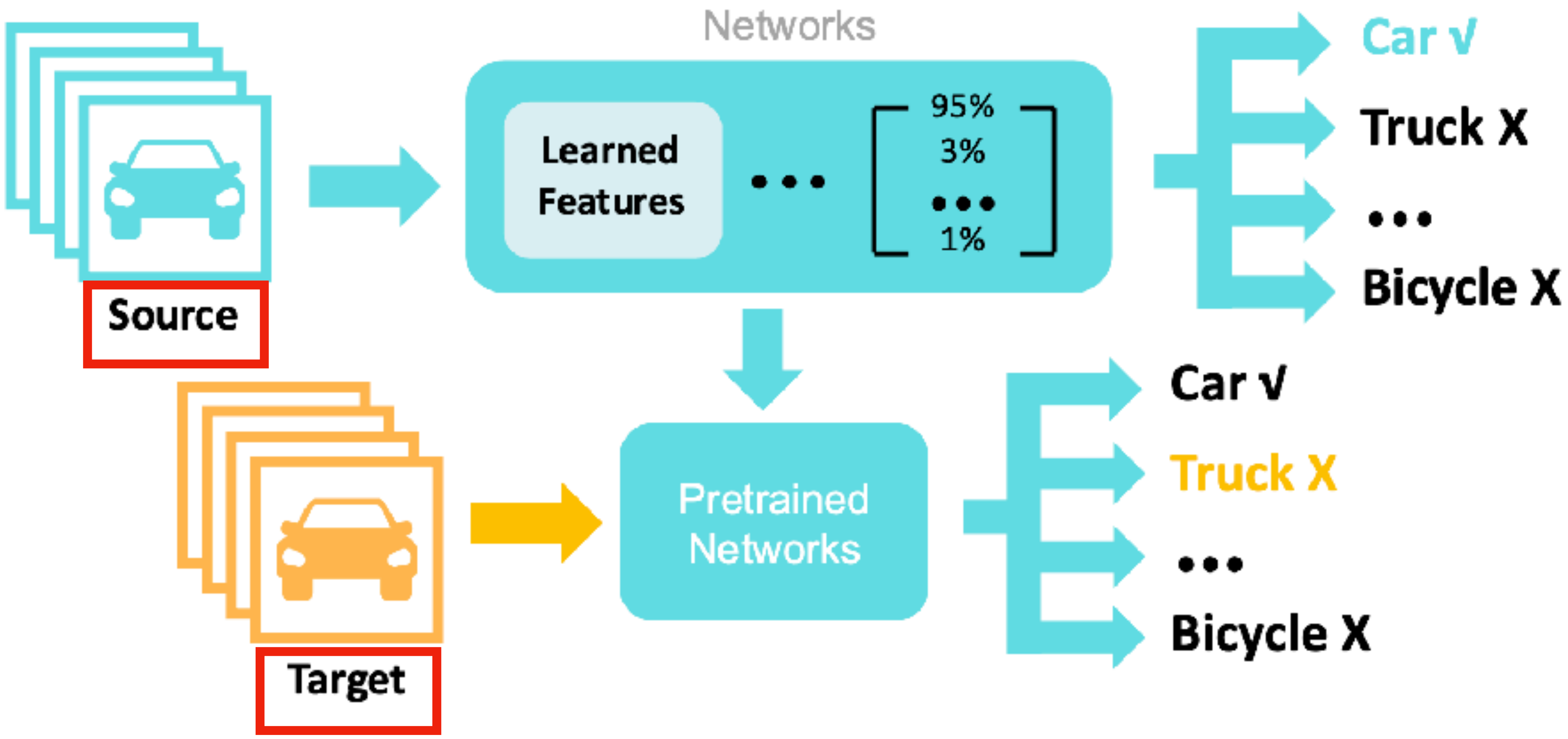
Transfer Learning Under Distribution Shift



Research Background — Distribution / Covariate Shifts



$$P_s(x) \neq P_t(x), P_s(y|x) = P_t(y|x)$$



Distribution shift may cause model performance degradation.

Lei Zhang et al. Transfer Adaptation Learning: A Decade Survey, *2022*.

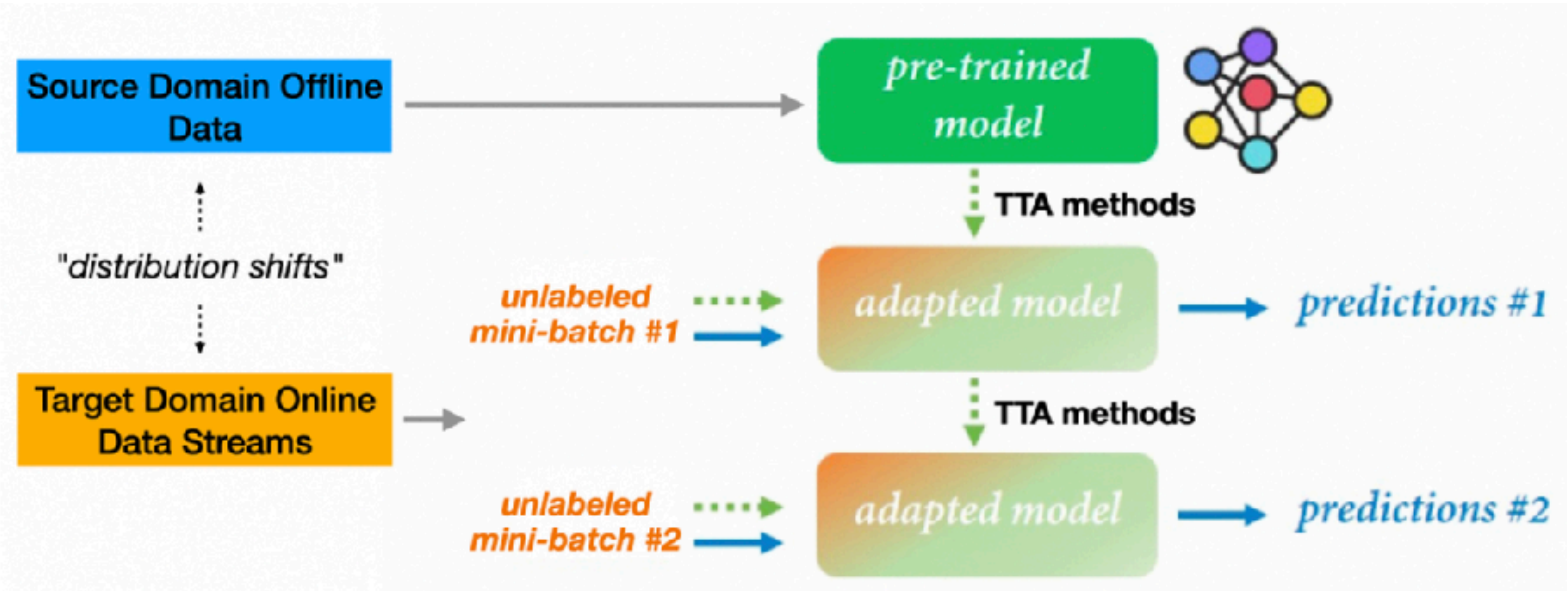
Fuzhen Zhuang et al. A Comprehensive Survey on Transfer Learning, *2020*.

Research Background — Test-time Adaptation (TTA)

TABLE I: Comparison of different transfer learning settings.

Topic	No Source Data*	No Target Labels*	Online Adaptation	Model Agnostic
Supervised Domain Adaptation	×	×	×	–
Unsupervised Domain Adaptation	×	✓	×	–
Source-free Domain Adaptation	✓	✓	×	–
Test-time Training	✓	✓	✓	×
Test-time Adaptation	✓	✓	✓	✓

Note: * denotes particular constraints during the adaptation process.



Given any pre-trained model, TTA aims to adapt it at test-time towards the unseen and unlabeled data streams.

Jian Liang, et. al. A Comprehensive Survey on Test-Time Adaptation under Distribution Shifts, *2024*.

Proposed Solutions x 3

How to safeguard DL models against distribution shifts?

- Sol#1: Test-time EMA for Small-batch Data Streams
- Sol#2: Un-mixing TTA for Heterogenous Data Streams
- Sol#3: Navigating Distribution Shifts in MedIA

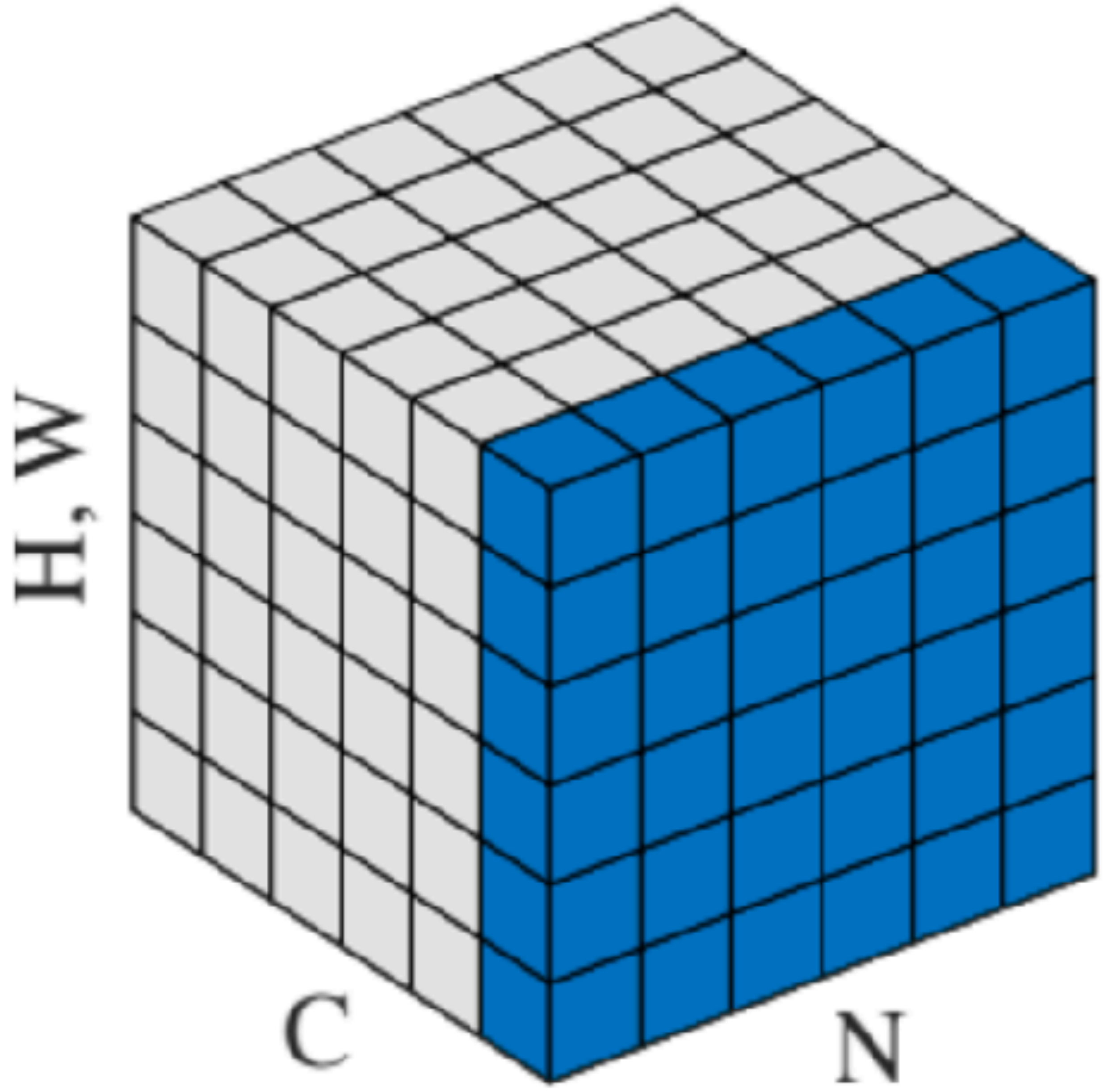
Solution #1:Test-time EMA for Small-batch Data Streams

How to safeguard DL models against distribution shifts?

- Sol#1: Test-time EMA for Small-batch Data Streams
- Sol#2: Un-mixing TTA for Heterogenous Data Streams
- Sol#3: Navigating Distribution Shifts in MedIA

Solution #1:Test-time EMA for Small-batch Data Streams

Batch Normalization in Domain Adaptation



$$\hat{\mathbf{x}}_i = \gamma \cdot \left(\frac{\mathbf{x}_i - \boxed{\boldsymbol{\mu}}}{\sqrt{\boxed{\boldsymbol{\sigma}^2} + \epsilon}}\right) + \beta$$

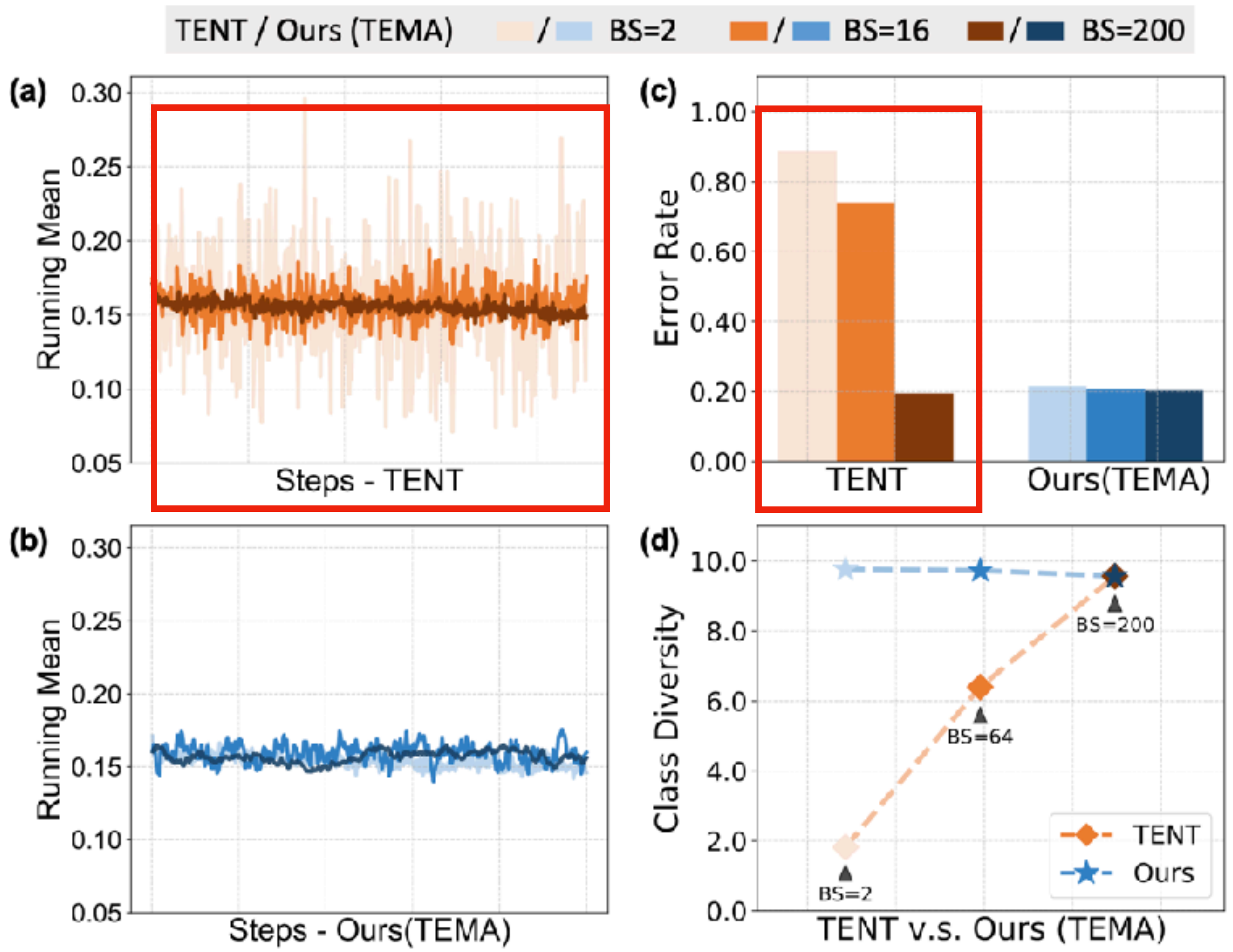
Parameters	How are they learned during training?	Testing
$\boldsymbol{\mu}, \boldsymbol{\sigma}$	exponential moving average with momentum m : $\boldsymbol{\mu} \leftarrow m\boldsymbol{\mu} + (1-m)\boldsymbol{\mu}_b$ $\boldsymbol{\sigma}^2 \leftarrow m\boldsymbol{\sigma}^2 + (1-m)\boldsymbol{\sigma}_b^2$ $\boldsymbol{\mu}_b = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i, \boldsymbol{\sigma}_b^2 = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \boldsymbol{\mu}_b)^2$	Fixed
γ, β	updated by gradients	Fixed

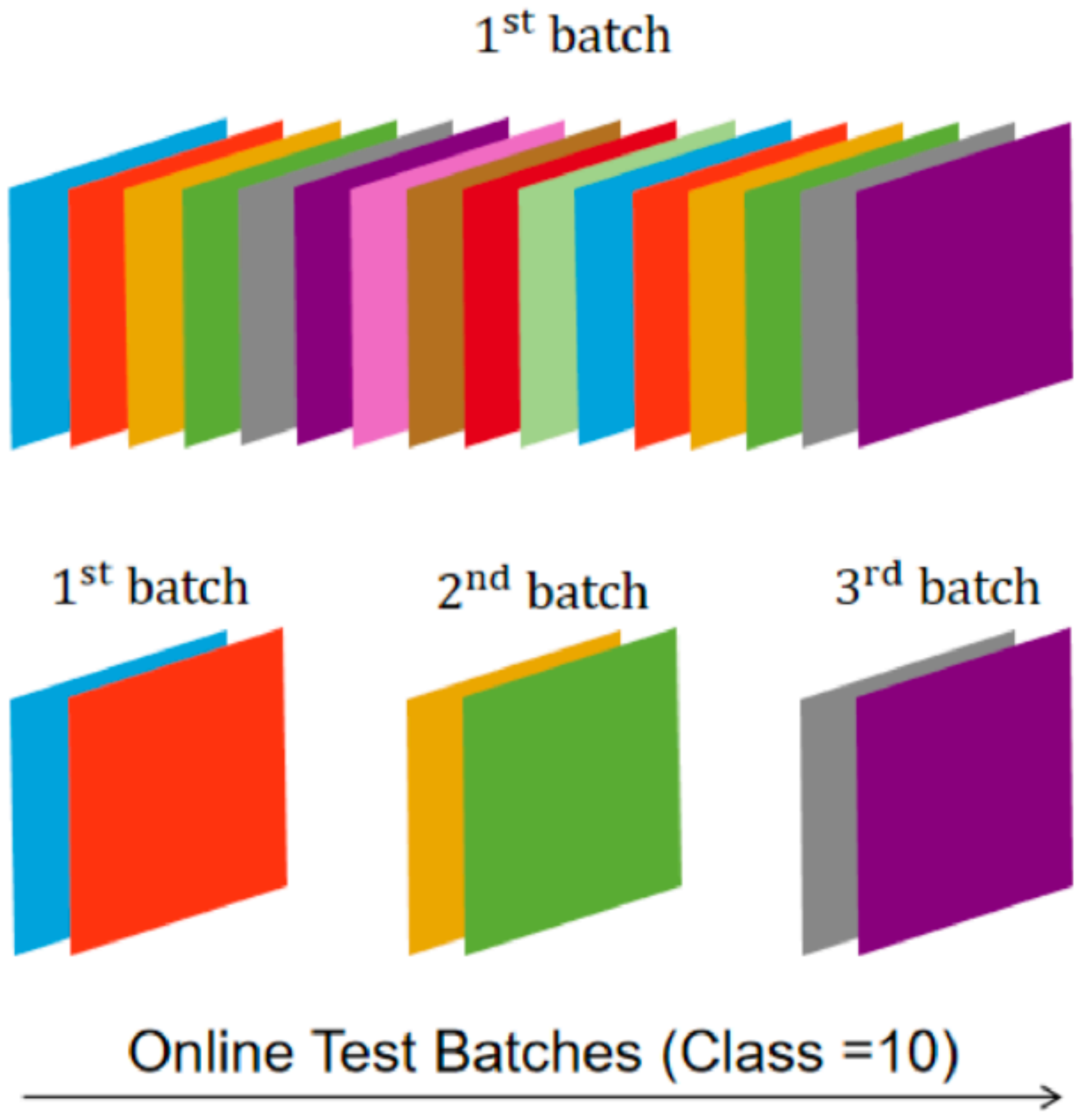
Normalization statistics are associated with domain characteristics.

Zixian Su, **Jingwei Guo**, et. al.Unraveling Batch Normalization for Realistic Test-Time Adaptation, *AAAI 2024*.

Solution #1:Test-time EMA for Small-batch Data Streams

Small-batch Model Degradation



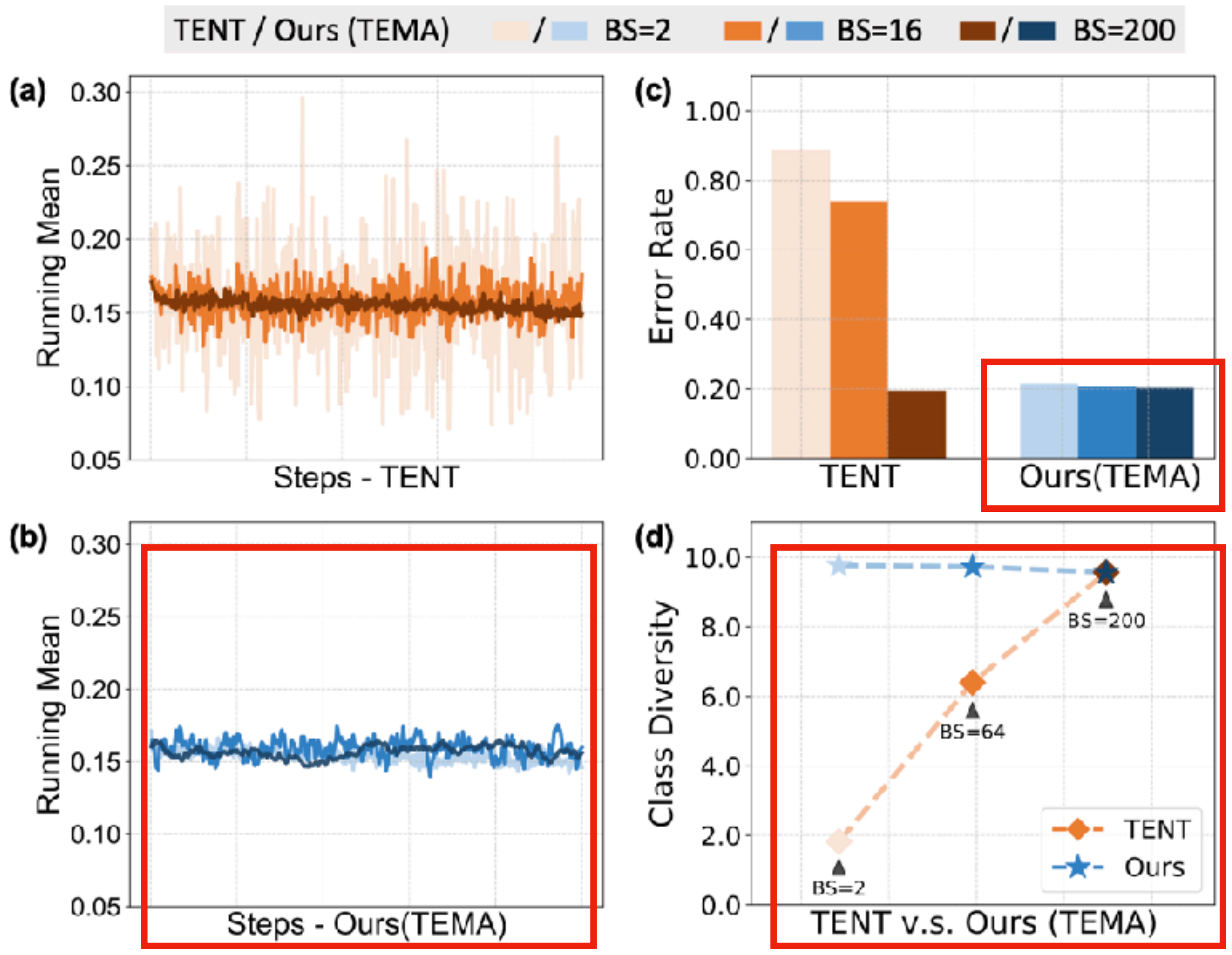


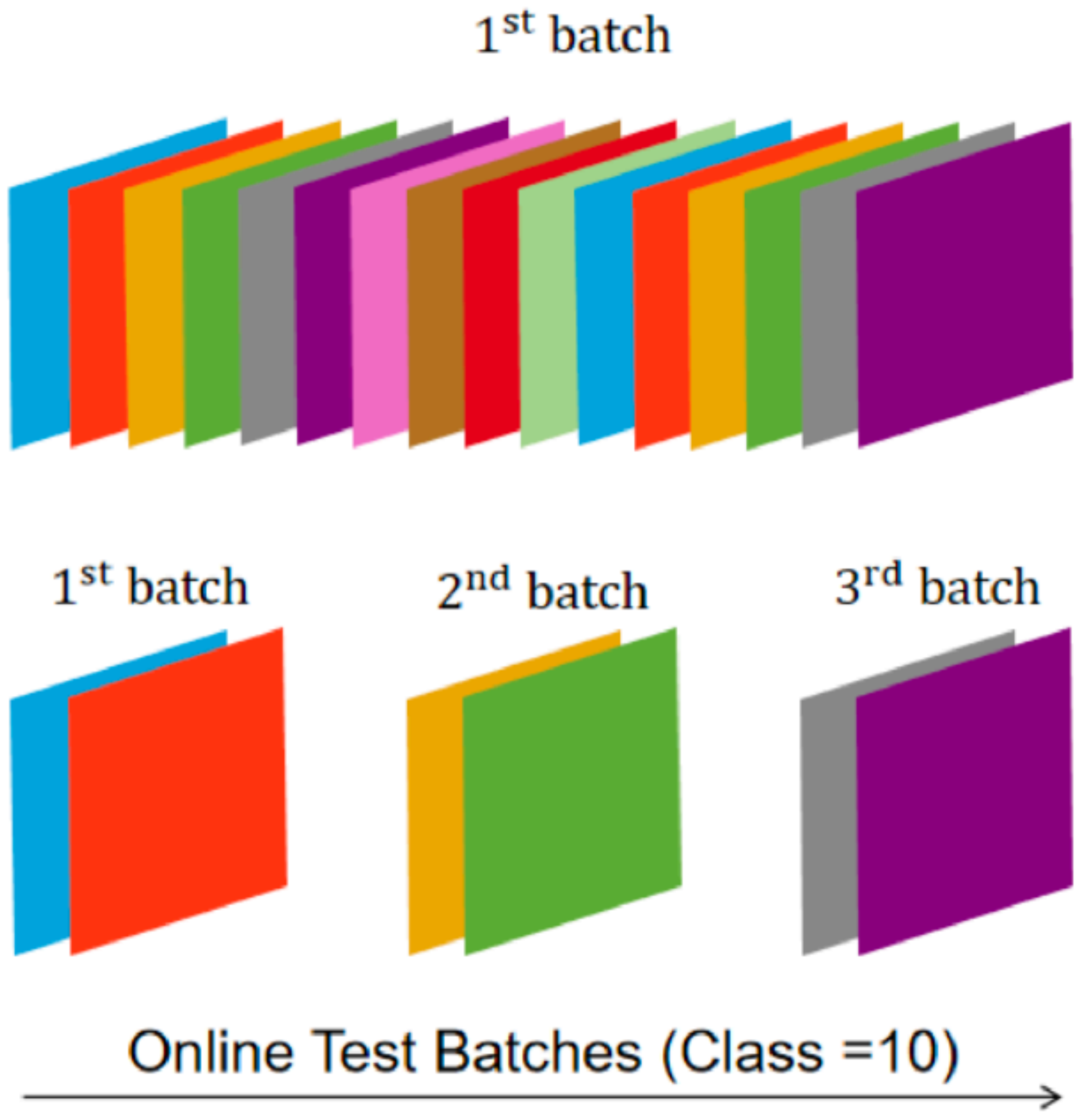
Traditional TTA methods tend to degrade on small-batch data streams.

Zixian Su, **Jingwei Guo**, et. al.Unraveling Batch Normalization for Realistic Test-Time Adaptation, *AAAI 2024*.

Solution #1:Test-time EMA for Small-batch Data Streams

Small-batch Model Degradation





TTA methods degrade on small batches due to reduced class diversity.

Zixian Su, **Jingwei Guo**, et. al.Unraveling Batch Normalization for Realistic Test-Time Adaptation, *AAAI 2024*.

Exponential Moving Average at Test-time

$$\mu_t^{ema} = m \cdot \mu_t^{batch} + (1-m) \cdot \mu_{t-1}^{ema}$$

$$\sigma_t^{ema2} = m \cdot \sigma_t^{batch2} + (1-m) \cdot \sigma_{t-1}^{ema2}$$

- Larger m emphasizes local batches
- Smaller m prioritize global statistics

Proposition 2. *Given the iterative rules of TEMA defined in Eq. 3 and 4, it yeilds the i-th term as a cumulative sum of the past batch statistics weighted by $w_0 = (1-m)^i$ for initial batch and $w_t = (1-m)^{(i-t)}m$ for $t = 1, 2, ..., i$. Let ϵ be a threshold defining the effective sample batch, such that only batches with a relative weight $w_t/w_i > \epsilon$ are included. Then, the expanded sample scope for statistical estimation in TEMA can be formally expressed as $\hat{N}_t = \lfloor \log_{1-m} \epsilon \rfloor \cdot N_t$.*

We apply grid search to tune m, guide by a tailored objective that balance local and global statistics.

$$\arg\min_m \boxed{\left| \frac{E(M_s|N_s)}{E(M_t|N_t)} - 1 \right|} + \boxed{\lambda \cdot \frac{N_t}{N_s}}$$

Aligning Class Diversity & Sample Scope

Zixian Su, **Jingwei Guo**, et. al.Unraveling Batch Normalization for Realistic Test-Time Adaptation, *AAAI 2024*.

Solution #1:Test-time EMA for Small-batch Data Streams

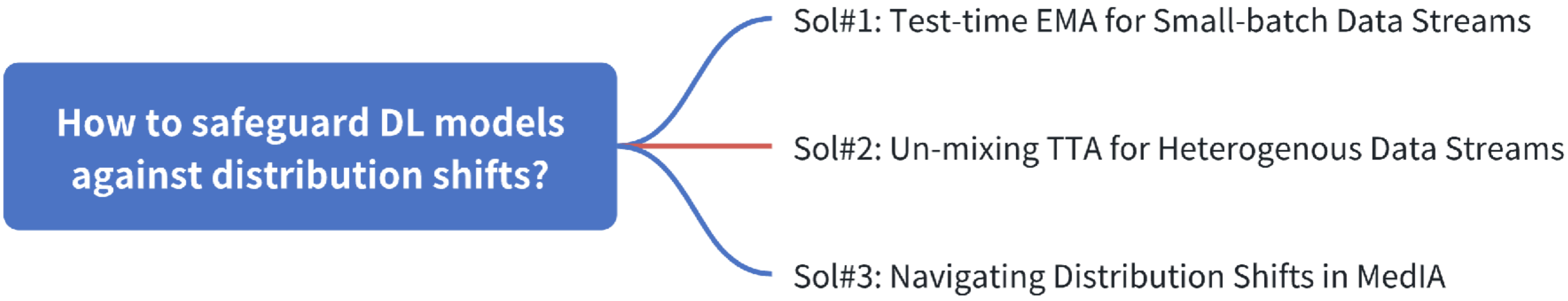
Empirical Results — Error Reduction

Continual		CIFAR-10-C 200	64	16	4	2	1	Avg.	CIFAR-100-C 200	64	16	4	2	1	Avg.
	Source	43.50	43.50	43.50	43.50	43.50	43.50	43.50	46.45	46.45	46.45	46.45	46.45	46.45	46.45
TR	TENT	19.55	26.32	74.07	85.07	88.69	90.00	63.95	61.12	86.49	96.07	98.40	98.79	99.00	89.98
TR	CoTTA	16.24	17.65	34.36	78.88	87.79	90.00	54.15	32.68	34.30	47.52	92.62	97.84	98.96	67.32
TR	SAR	20.40	20.74	22.89	31.35	40.32	89.83	37.59	31.90	35.89	54.84	66.08	73.24	98.91	60.14
TR	AdaCont.	18.50	17.41	19.69	35.01	63.81	31.55	31.00	33.61	35.40	55.04	89.45	96.16	62.67	62.06
TR	ETA	17.64	20.01	30.60	56.78	83.24	89.83	49.68	32.31	35.17	44.72	88.22	98.96	98.91	66.38
TF	TBN	20.35	20.82	23.06	31.62	38.57	89.83	37.38	35.50	36.29	39.67	52.73	73.24	98.91	56.06
TF	α-BN	30.60	30.67	30.89	31.89	32.91	34.47	31.91	37.02	38.27	35.92	37.17	37.25	41.18	37.80
TF	AdaptBN	20.36	20.71	21.98	26.79	32.19	37.52	26.59	35.40	**35.78**	35.92	37.14	39.23	41.85	37.55
TF	LAME	64.52	57.66	47.70	44.38	43.75	90.00	58.00	98.49	73.83	47.59	46.64	46.50	99.00	68.68
TF	**Ours**	**20.20**	**20.57**	**20.74**	**21.45**	**20.91**	**21.05**	**20.82**	**34.63**	36.11	**35.31**	**36.02**	**36.32**	**39.30**	**36.28**

Table 1: Continual adaptation on corruption benchmark CIFAR-10-C/CIFAR-100-C. Error rate (↓) averaged over 15 corruption with severity level 5 for each test batch size (200/64/16/4/2/1).

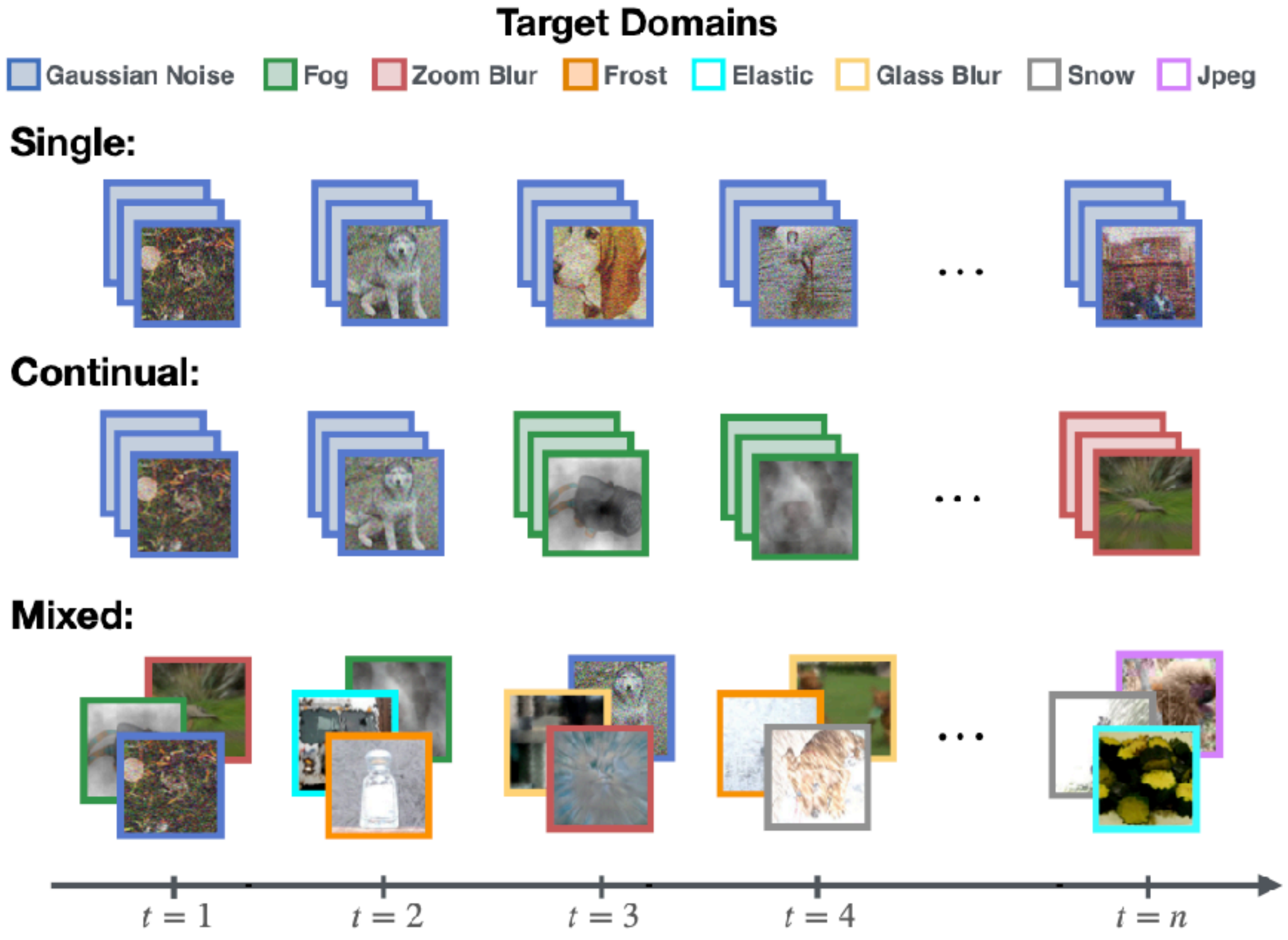
Zixian Su, **Jingwei Guo**, et. al.Unraveling Batch Normalization for Realistic Test-Time Adaptation, *AAAI 2024*.

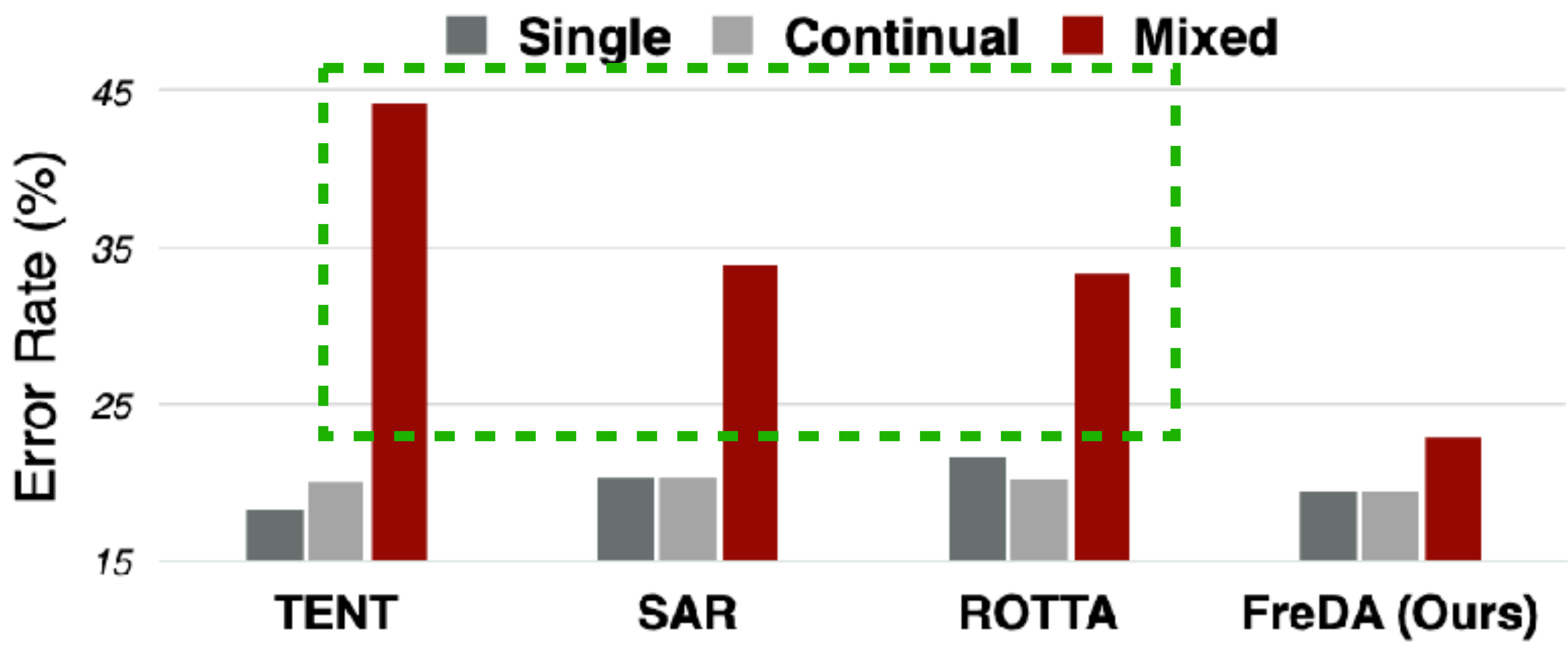
Solution #2: Un-mixing TTA for Heterogeneous Data Streams



Solution #2: Un-mixing TTA for Heterogeneous Data Streams

Heterogeneous / Mixed-domain Model Degradation



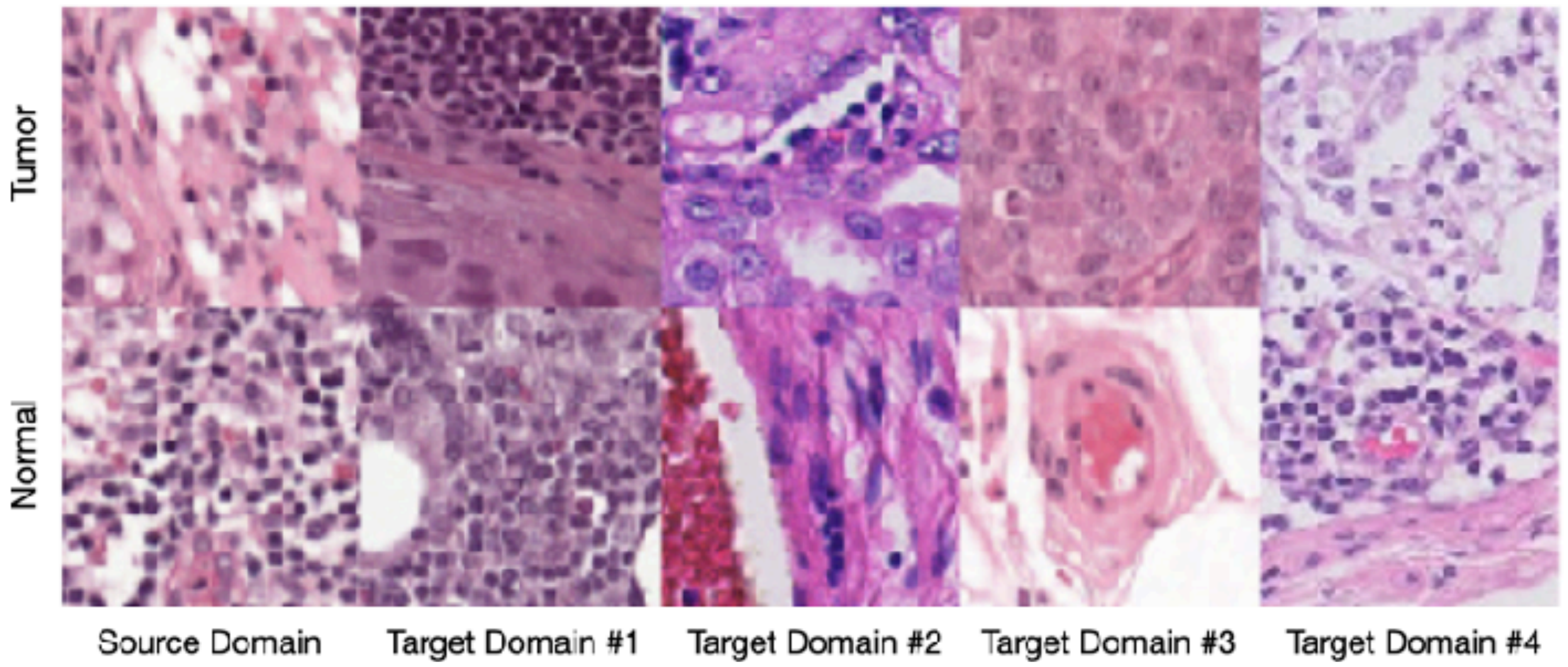


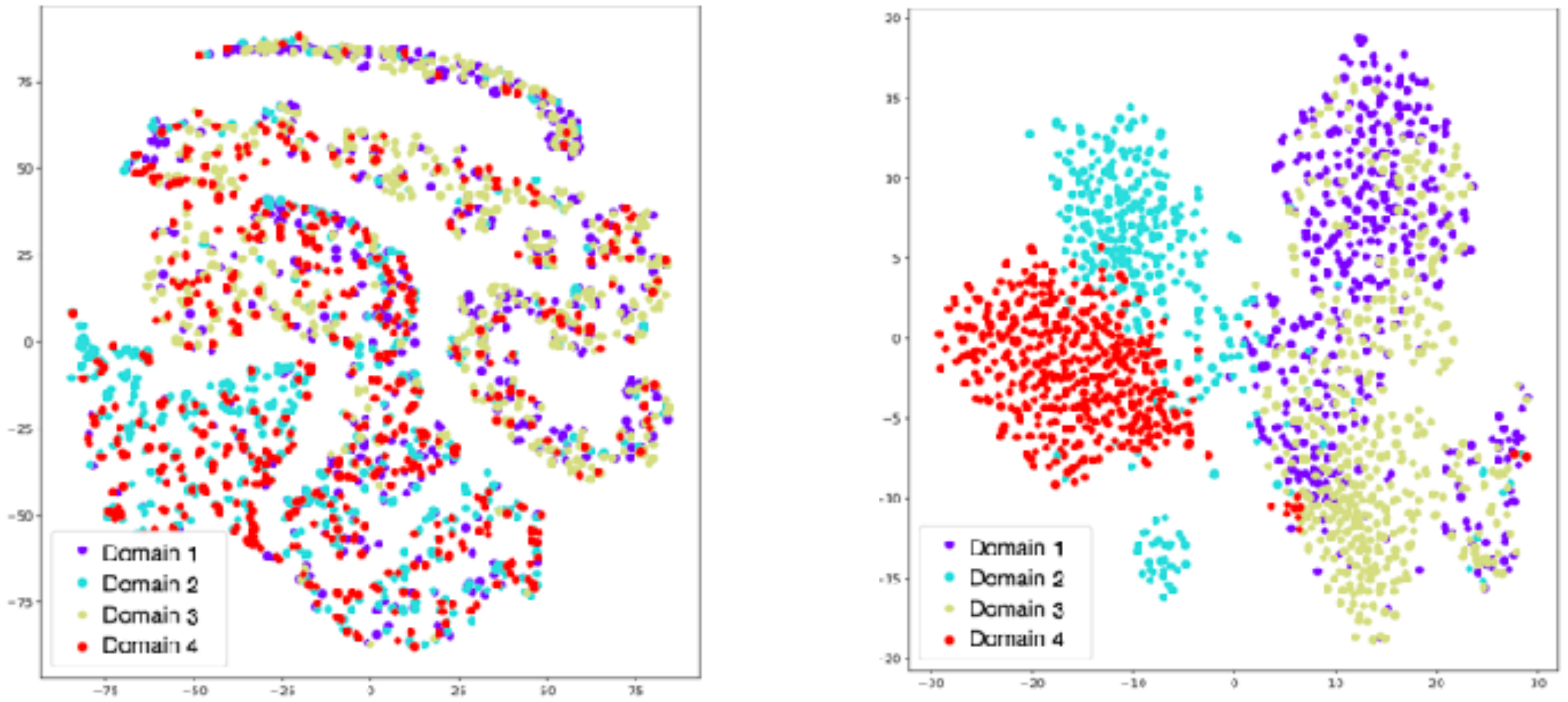
Conventional TTA methods assume homogenous data streams and employ whole-batch adaptation strategies.

Zixian Su*, **Jingwei Guo***, et. al., Un-mixing Test-time Adaptation under Heterogeneous Data Streams, *TKDE (Under Review) 2025*.

Solution #2: Un-mixing TTA for Heterogeneous Data Streams

Frequency-based Domain Separation



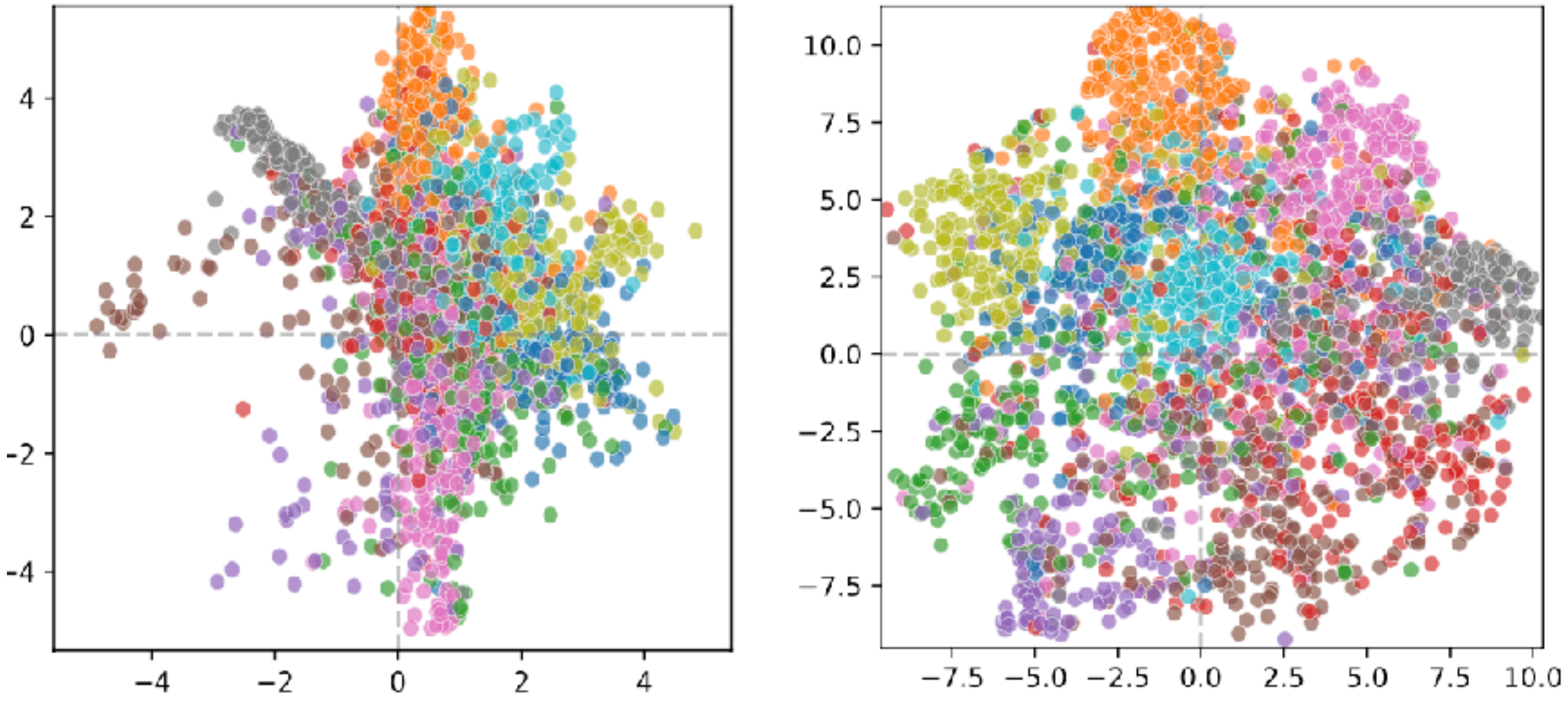


High-frequency information enable target subdomains' separation, contrasting with pre-trained model features.

Error rate (%)

Methods	C10	C100	IN
TBN (Centralized)	33.8	45.8	82.5
TBN (Decentralized)	28.5	43.2	77.6

Class Separation

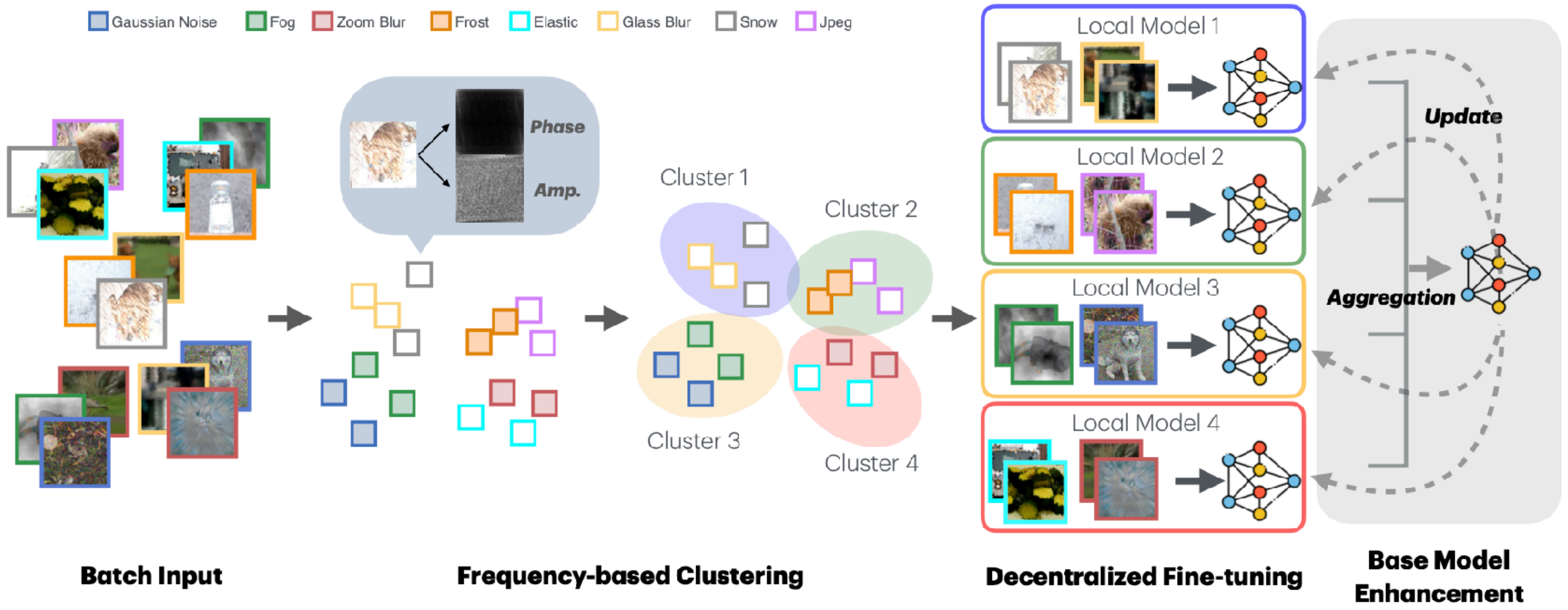


(a) Centralized Adaptation (b) Decentralized Adaptation

Zixian Su*, **Jingwei Guo***, et. al., Un-mixing Test-time Adaptation under Heterogeneous Data Streams, *TKDE (Under Review) 2025*.

Solution #2: Un-mixing TTA for Heterogeneous Data Streams

Overall Framework



Zixian Su*, **Jingwei Guo***, et. al., Un-mixing Test-time Adaptation under Heterogeneous Data Streams, *TKDE (Under Review) 2025*.

Solution #2: Un-mixing TTA for Heterogeneous Data Streams

Empirical Results

Baseline & Methods	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brig.	Contr.	Elast.	Pixel	JPEG	Avg.
CIFAR-10-C (WRN-28)	72.3	65.7	72.9	46.9	54.3	34.8	42.0	25.1	41.3	26.0	9.3	46.7	**26.6**	58.4	30.3	43.5
TBN	45.5	42.8	59.7	34.2	44.3	29.8	32.0	19.8	21.1	21.5	9.3	27.9	33.1	55.5	30.8	33.8
TENT (ICLR 21')	73.5	70.1	81.4	31.6	60.3	29.6	28.5	30.8	35.3	25.7	13.6	44.2	32.6	70.2	34.9	44.1
ETA (ICML 22')	36.2	33.3	52.3	22.9	38.9	22.4	20.5	19.5	19.7	20.4	11.3	35.4	**26.6**	38.8	25.1	28.2
AdaContrast (CVPR 22')	36.7	34.3	48.8	**18.2**	39.1	21.1	17.7	18.6	18.3	16.8	**9.0**	17.4	27.7	44.8	**24.9**	26.2
CoTTA (CVPR 22')	38.7	36.0	56.1	36.0	**36.8**	32.3	31.0	19.9	**17.6**	27.2	11.7	52.6	30.5	35.8	25.7	32.5
SAR (ICLR 23')	45.5	42.7	59.6	34.1	44.3	29.7	31.9	19.8	21.1	21.5	9.3	27.8	33.0	55.4	30.8	33.8
RoTTA (CVPR 23')	60.0	55.5	70.0	23.8	44.1	20.7	21.3	20.2	22.7	**16.0**	9.4	22.7	27.0	58.6	29.2	33.4
RDumb (NeurIPS 23')	34.9	32.3	49.4	23.3	38.2	23.3	20.7	19.9	19.3	20.7	11.2	29.3	26.7	41.5	25.2	27.7
DeYO (ICLR 24')	45.8	42.3	65.7	21.3	41.8	25.1	19.5	21.1	19.6	19.2	12.3	21.8	28.5	39.3	28.0	30.1
UnMix-TNS (ICLR 24')	50.0	44.4	44.3	34.4	48.2	32.7	30.0	35.5	35.9	47.5	28.1	38.7	43.9	40.0	43.3	39.8
FreDA (ours)	**23.1**	**22.2**	**32.2**	18.7	41.6	**18.8**	**16.8**	**17.9**	19.9	16.9	9.8	**13.2**	29.1	**35.4**	28.6	**22.9**
CIFAR-100-C (ResNeXt-29)	73.0	68.0	39.4	29.3	54.1	30.8	28.8	39.5	45.8	50.3	29.5	55.1	37.2	74.7	41.2	46.4
TBN	62.7	60.7	43.1	35.5	50.3	35.7	34.4	39.9	51.5	**27.5**	45.5	42.3	72.8	46.4	45.8	45.8
TENT (ICLR 21')	95.6	95.2	89.2	72.8	82.9	74.4	72.3	78.0	79.7	84.7	71.0	88.5	77.8	96.8	78.7	82.5
ETA (ICML 22')	42.6	40.3	**34.1**	30.3	42.4	32.0	29.4	35.6	35.8	44.1	30.2	41.8	**36.9**	38.9	40.9	37.0
AdaContrast (CVPR 22')	54.5	51.5	37.6	30.7	45.4	32.1	30.3	36.9	36.5	45.3	28.0	42.7	38.2	75.4	41.7	41.8
CoTTA (CVPR 22')	54.4	52.7	49.8	36.0	45.8	36.7	33.9	38.9	35.8	52.0	30.4	60.9	40.2	38.0	41.1	43.1
SAR (ICLR 23')	75.8	72.7	41.1	**29.2**	45.2	31.1	28.9	36.7	37.7	43.9	29.3	41.8	37.1	89.2	42.4	45.5
RoTTA (CVPR 23')	65.0	62.3	39.3	33.4	50.0	34.2	32.6	36.6	36.5	45.0	**26.4**	41.6	40.6	89.5	48.5	45.4
RDumb (NeurIPS 23')	42.3	40.0	**34.1**	30.5	42.4	31.9	29.5	35.7	35.9	43.6	30.4	41.9	36.9	38.1	40.5	36.9
DeYO (ICLR 24')	57.2	53.4	38.8	34.7	47.3	37.3	34.1	40.8	40.5	50.6	33.3	45.8	41.5	94.5	45.7	46.4
UnMix-TNS (ICLR 24')	65.8	64.1	46.4	37.5	51.7	36.0	36.4	38.5	39.4	51.1	29.3	42.8	43.2	67.8	49.4	46.6
FreDA (ours)	**34.8**	**34.7**	36.6	29.4	**41.2**	**29.9**	**28.4**	**33.8**	**33.7**	41.1	29.8	**34.9**	**36.9**	**37.1**	**38.7**	**34.7**
IN-C (ResNet-50)	97.8	97.1	98.2	81.7	89.8	85.2	77.9	83.5	77.1	75.9	41.3	94.5	82.5	79.3	68.6	82.0
TBN	92.8	91.1	92.5	87.8	90.2	87.2	82.2	82.2	82.0	79.8	48.0	92.5	83.5	75.6	70.4	82.5
TENT (ICLR 21')	99.2	98.7	99.0	90.5	95.1	90.5	84.6	86.6	84.0	86.5	46.7	98.1	86.1	77.7	72.9	86.4
ETA (ICML 22')	90.7	89.2	90.5	77.0	**80.6**	74.0	68.9	72.4	70.3	64.6	43.9	93.4	**69.2**	**52.3**	**55.9**	72.9
AdaContrast (CVPR 22')	96.2	95.5	96.2	93.2	96.4	96.3	90.5	92.7	91.9	92.4	50.8	97.0	96.6	89.7	87.1	90.8
CoTTA (CVPR 22')	89.1	86.6	88.5	80.9	87.2	81.1	75.8	73.3	75.2	70.5	41.6	85.0	78.1	65.6	61.6	76.0
SAR (ICLR 23')	98.4	97.3	98.0	84.0	87.3	82.6	77.2	77.5	76.1	72.5	43.1	96.0	78.3	61.8	60.4	79.4
RoTTA (CVPR 23')	89.4	88.6	89.3	83.4	89.1	86.2	80.0	78.9	76.9	74.2	**37.4**	89.6	79.5	69.0	59.6	78.1
RDumb (NeurIPS 23')	89.0	87.6	88.6	78.1	82.3	75.2	70.1	73.0	71.0	65.1	43.9	92.6	70.7	53.7	56.3	73.1
DeYO (ICLR 24')	99.5	99.2	99.5	89.5	95.0	83.9	78.8	75.0	87.8	79.2	47.3	99.2	92.4	59.0	60.4	83.0
UnMix-TNS (ICLR 24')	91.7	92.8	91.7	92.3	93.4	91.5	84.8	86.3	84.1	85.0	62.0	96.5	88.6	81.7	77.3	86.7
FreDA (ours)	**72.4**	**74.0**	**71.4**	**76.5**	82.3	**72.1**	**64.1**	**64.4**	**64.8**	**59.1**	43.7	**79.7**	71.0	54.2	58.6	**67.2**

Error Rate -3.73%
under Mixed
Distribution Shifts

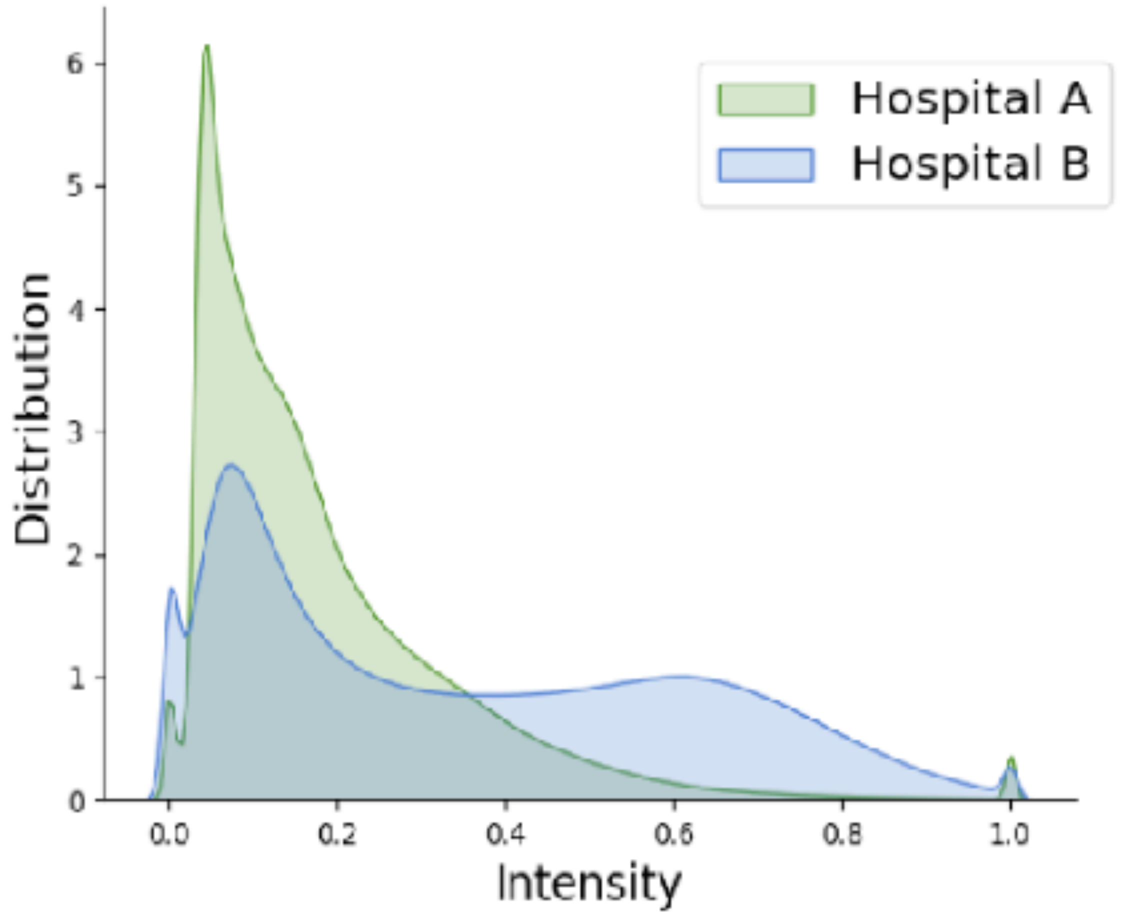
Zixian Su*, **Jingwei Guo***, et. al., Un-mixing Test-time Adaptation under Heterogeneous Data Streams, *TKDE (Under Review) 2025*.

Solution #3: Navigating Distribution Shifts in MedIA: A Survey

How to safeguard DL models against distribution shifts?

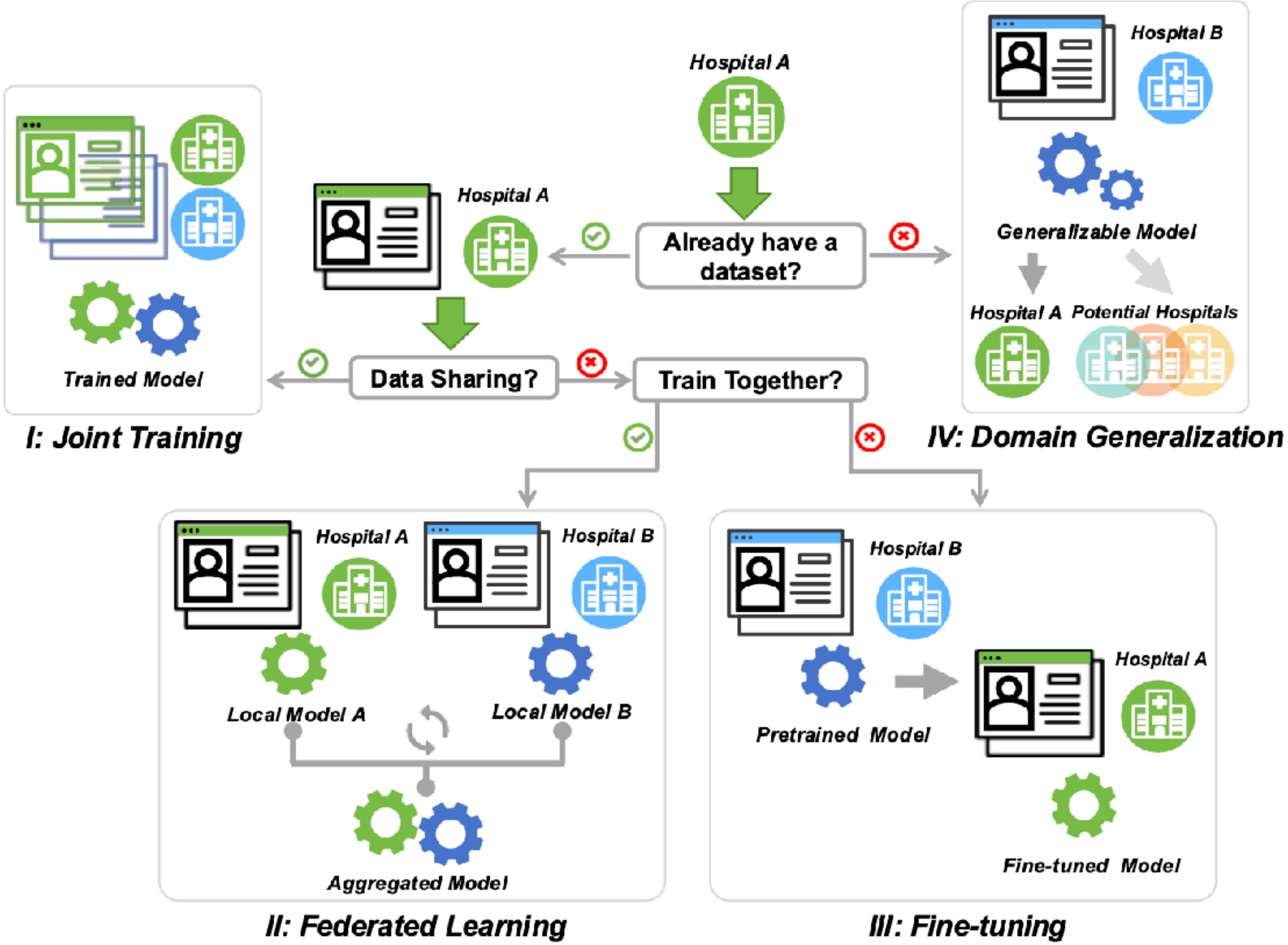
- Sol#1: Test-time EMA for Small-batch Data Streams
- Sol#2: Un-mixing TTA for Heterogenous Data Streams
- Sol#3: Navigating Distribution Shifts in MedIA

Solution #3: Navigating Distribution Shifts in MedIA: A Survey



Suppose Hospital A (target) seeks a model tailored for its distribution, developed in collaboration with Hospital B (source):

- Data Accessibility
- Privacy Concerns
- Collaborative Protocols



Zixian Su*, **Jingwei Guo***, et.al. Navigating Distribution Shifts in Medical Image Analysis. *ACM Comput. Surv (Under Review) 2025.*

Research Summary